Robust modelling of battery revenues and economic viability in resource adequacy studies - Methodology Report

17 December 2025

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Executive Summary

Enhanced framework integrating key Revenue streams with market-based validation

- The project delivers a comprehensive methodology for TenneT to evaluate utility-scale battery revenues and business cases consistent with TenneT's MLZ Report. The methodology builds on TenneT's existing framework for day-ahead modeling, with supplementary elements implemented in accessible tools (python script and excel files) using a workflow that can be applied consistently across all study years.
- The approach integrates relevant revenue streams—day-ahead markets, intraday markets, balancing capacity, and ancillary services—through a structured workflow that combines insights from PLEXOS fundamental market analysis with Python-based tools for intraday price synthesis and revenue optimization.
- The Python-based sampling technique generates quarter-hourly intraday price series that maintains arbitrage-free consistency with day-ahead simulations while capturing the higher volatility characteristic of residual markets. A rescale factor based on historical correlations ensures that fundamental changes in day-ahead markets are appropriately reflected in intraday valuations, maintaining the link to market fundamentals. This proportional relationship reflects historical analysis showing intraday spreads are typically 2-3x higher than day-ahead spreads.
- Reserve provision is calculated within the PLEXOS day-ahead model, with exogenous pricing based on opportunity cost logic linked to day-ahead and intraday modeling results. These reserve commitments are passed through as constraints to the intraday optimization, preventing double allocation of capacity.
- The methodology addresses revenue stacking opportunities while recognizing participation constraints through sequential optimization logic that prevents double counting and reflects market depth.

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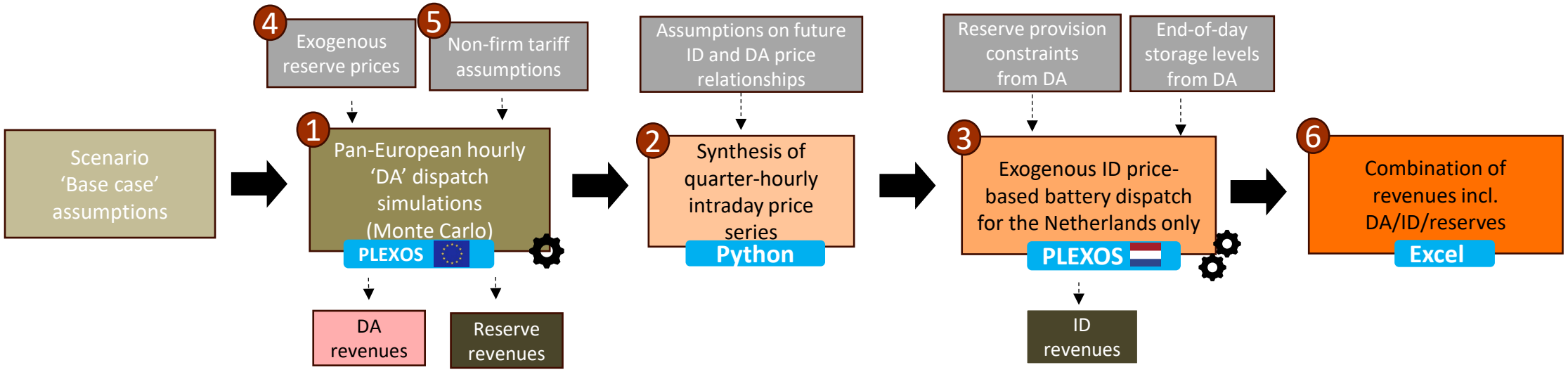
Overview of proposed solution approach for BESS revenues

Leveraging PLEXOS model for BESS revenue evaluations:

- The overall proposed solution ensures that it covers the value driver approach of TenneT's current methodology. It includes the BESS intrinsic value which can be derived from the electricity day-ahead market, as well as the additional value which can be derived from the intraday market and from the ancillary service markets. In the following slides, we present in more details each of the steps of the workflow below and their underlying assumptions.



System cost minimisation algorithm

Asset profit maximisation algorithm

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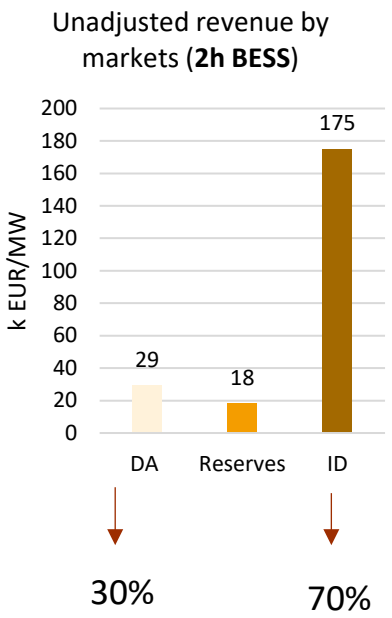
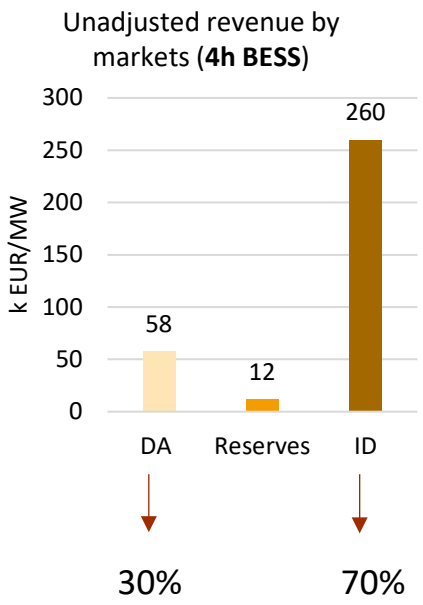
Results for individual market segments and stacked revenues

Results from a single price scenario for the year 2030 for an average climate year

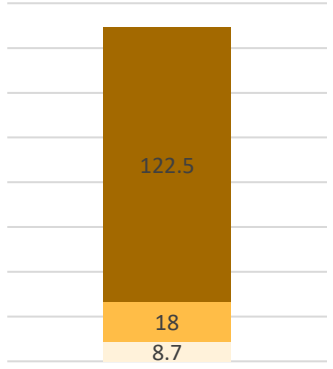
- **The results showed in the bar charts on the right are underlined by:**
- Synthetic ID price series rescaled by a factor of 0.85.
- Reserve provision allocated to BESS from the DA model output, as well as EOD battery storage level, are passed on into the ID optimization.
- Included a VOM cost of 10 EUR/MWh for intraday optimization.
- Exogeneous upward reserve prices of 12 EUR/MW/h and downward price of 5 EUR/MW/h and a limit to BESS reserve provision of 50% of the asset capacity.
- BESS are allowed to provide FCR and aFRR, both in upward and downward directions, but are excluded from mFRR provision.

- **The figures on the right show the results based the recommended split of BESS capacity allocation between the day-ahead (30%) and the intraday market (70%).**
- This weighting factor is based on KYOS's modeling results, which support asset co-optimization across both markets. It might place slightly more emphasis on the day-ahead market. It helps address the risk that intraday markets may lack sufficient liquidity to trade all BESS capacity, making it a cautious and reasonable approach.
- The revenues from reserve provision can be added to the revenue streams because the approach ensures that double-counting is avoided.

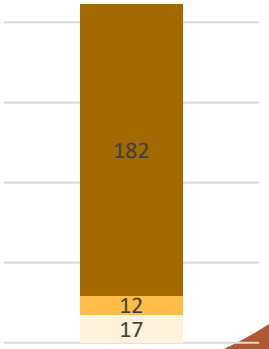




149 kEUR/MW/year



211 kEUR/MW/year



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Introduction & Background

Primary Goals of the Security of Supply Monitor (MLZ) report

- TenneT's annual MLZ report analyses the expected security-of-supply developments in the Netherlands over the coming ten years. It is part of TenneT's legal mandate is to monitor the security of supply on an annual basis, and deliver its assessment and advice to the Dutch government.
- As part of the MLZ, TenneT performs an economic viability assessment (EVA) which assesses the viability of groups of capacity resources based on the difference between expected revenues and costs.

Challenges with the EVA assessment of Battery Energy Storages Systems (BESS)

- In the EVA, future power plant electricity market revenues are estimated based on the results of the hourly unit commitment and economic dispatch (UCED) simulations of the electricity market, combined with additional revenue streams computed ex-post outside the model (e.g. ancillary services revenues) to arrive at the total plant value.
- This former approach, which emphasized day-ahead value with ex-post adjustments, was deemed insufficient for BESS revenue assessments as it could overlook price variations, volatility, traded volumes, revenue stacking from ancillary services, and effects of non-firm network agreements and time-of-use tariffs.

Project Objectives

- KYOS was asked to analyse TenneT's approach to BESS revenue modeling and recommend a methodology to thoroughly evaluate BESS revenues, consistent with TenneT's MLZ Report and the UCED market simulation framework.
- The work content of this report summarises the three phases of the project:
  - A review of TenneT's existing approach to BESS revenue assessments
  - The proposed battery valuation methods
  - Test case results and validation of the proposed methodology

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The main steps of this project

The project consisted of four main steps outlined below, and included the delivery of two reports.

Overview of BESS economics

- An overview of battery economics and market participation considering not only the current situation but also expected trends in key value drivers, main business models, trading behaviour, typical distribution of revenues across different markets, and investment decisions.
- This is covered in a separate report, shared next to this one.

Review of existing approach

- A review of the extent to which the BESS economics outlined in the "Battery Economics and Landscape report" is captured in TenneT's existing modeling approach.
- A comparison between TenneT and KYOS modelling approaches and drawing conclusions for the development of a new methodology.

Proposed Method

- Development and implementation of a robust (yet simplified) methodology for estimating the revenues of batteries from all relevant sources (e.g. day-ahead markets, intraday markets, balancing capacity, balancing energy,…) and overall viability of batteries on the basis of electricity market simulation results.

Validation

- Validation of the developed methodology to demonstrate its robustness and quantify its accuracy against KYOS battery revenue modelling results.

Battery Economics and Landscape Report

Robust modelling of battery revenues and economic viability in resource adequacy studies – Methodology Report

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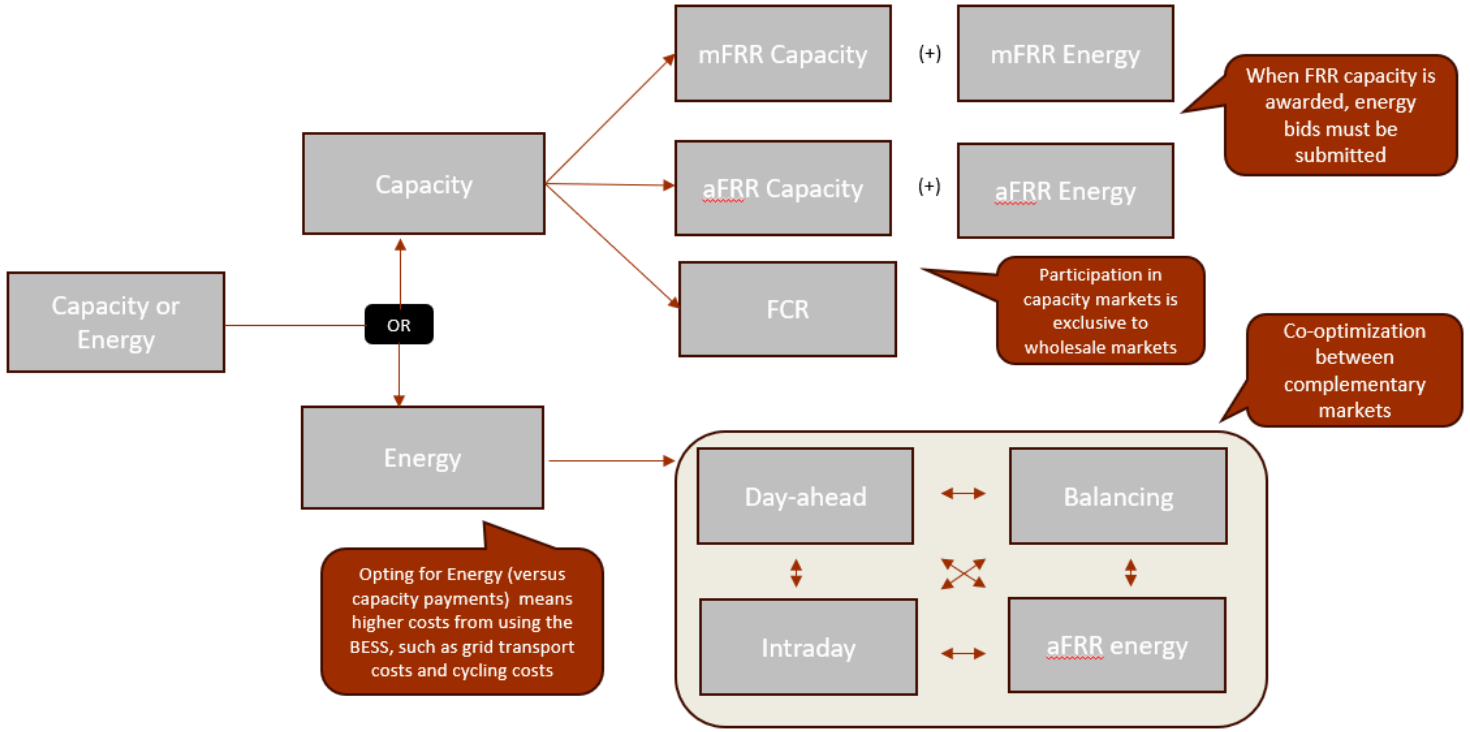
Approach and principles used in this project

The proposed approach should be comprehensive and internally consistent with how market parties engage with the different market opportunities

- The project centers on revenue modeling of utility-scale BESS, connected to the high-voltage (HV) and EHV grids.
- The methodology created must be comprehensive, covering the entire range of revenues from wholesale and balancing markets. The report will identify any revenue types that have not been modeled explicitly, for example where they were considered immaterial.
- It should consider opportunities for revenue stacking while recognizing participation constraints and the compromises between different revenue streams.
- Revenues from congestion management services, such as those from programs like GOPACS, are excluded.
- Non-market expenses, including grid fees, are not included in the proposed BESS revenue approach. However, it is included in the economic viability assessment itself.

Several revenue streams exist, although certain options cannot be pursued simultaneously



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Approach and principles used in this project

The proposed approach should be based on the core outputs of TenneT's fundamental model

- The EVA focuses on evaluating the economic viability of the supply mix within market environments rather than providing precise valuations at the individual asset level.
- Therefore, the approach must balance modeling accuracy with practical usability. The proposed method should address missing data while reducing complexity to deliver dependable results without unnecessary detail.
- It should remain based on TenneT's core day-ahead market modeling using PLEXOS.
- Supplementary elements should be readily accessible to TenneT, avoiding black-box solutions, and ideally implemented in tools like Excel or Python.

Anticipated process		Project Scope
Scenario 'Base case' assumptions	• Fuel & carbon prices • Installed capacities • Load • Etc.	• **Recommend enhancements to improve the accuracy of PLEXOS modeling.**
Hourly (~day-ahead) dispatch simulations (Monte Carlo) PLEXOS	• Hourly (optimized) dispatch and prices for ~360 possible realisations of future 'Base Case' scenario (36 climate scenarios x 10 forced outage patterns)	• **Suggest enhancements for day-ahead modeling**
Addition of other value drivers to business cases Python/Excel	• Optionality value on hourly shaping (additional extrinsic value which can be derived from intraday market) • Ancillary services	• **Propose methodology improvements for intraday and ancillary services**

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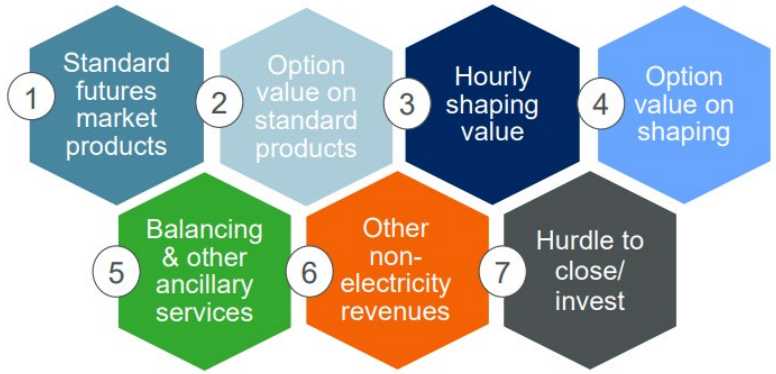
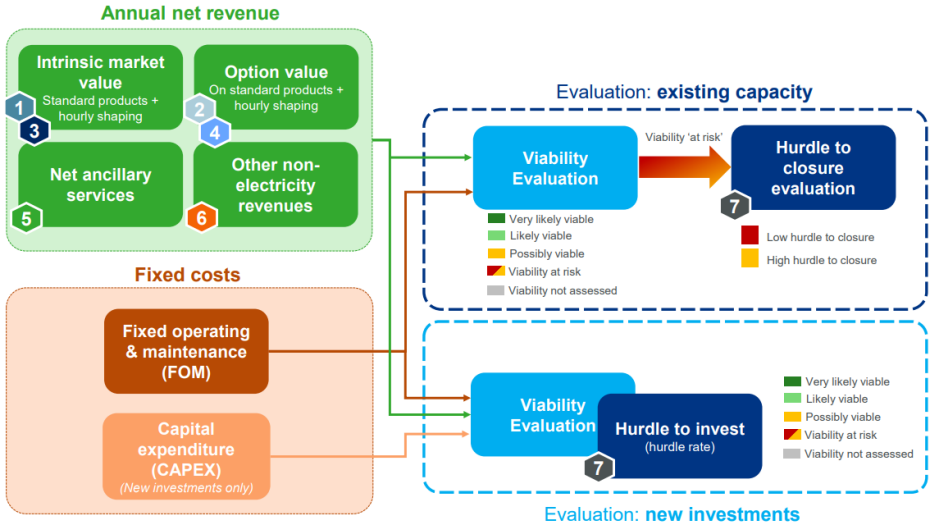
TenneT's existing approach (1)

- TenneT's approach to performing economic viability assessment (EVA) of batteries and other technologies (as explained in its whitepaper) is based on quantifying several types of revenues or value drivers (VDs) using an electricity market model, and comparing these modelled revenues with expected fixed costs.
- **Intrinsic market value** is the sum of *Value from standard futures products* (VD1) and *Hourly shaping value* (VD3). This represents the total intrinsic value which can be derived from the electricity futures and (day-ahead) spot markets.
  - *These drivers are estimated based on hourly simulations of a future scenario(s) of the pan-EU electricity market with a market modelling tool (PLEXOS) based on unit commitment and economic dispatch (UCED).*
- **Option value** is the sum of *Option value on standard futures products* (VD2) and *Option value on hourly shaping* (VD4), representing the additional extrinsic value which can be derived from futures/forwards and subsequent markets closer to real time.
  - *These are estimated using market data and Python tools outside the market model.*
- **Other value** from *Balancing & other ancillary services* (VD5) and *Other non-electricity revenues* (VD6) such as heat and stream is also estimated.
  - *These are also typically estimated outside the market model, independent of the dispatch in the market model.*

TenneT's Value Driver (VD) framework for performing EVA





Source: TenneT 2024, Analysing the economic viability of capacity resources for resource adequacy studies

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TenneT's existing approach (2)

TenneT identified the need to better reflect revenue streams realistically accessible to BESS

- Since performing the EVA assessment for battery technologies in the MLZ 2024 (and again in the MLZ 2025), TenneT identified that its electricity market modelling framework may not fully reflect all revenue streams realistically accessible to batteries. To address this, TenneT applied **two correction factors** to better approximate battery revenues that are not yet fully captured in the current modelling setup, as explained in the MLZ 2024 report annex:

  - **Volume correction**. TenneT applies a volume uplift of roughly 30% to scale modelled battery dispatch to a minimum of around 1.4 cycles/day (~500 cycles/year). Without this correction, the hourly market simulations result in approximately 1.1 cycles/day, which TenneT considers lower than expected for a battery actively participating in both day-ahead and intraday markets. The uplift therefore serves as a proxy for additional cycling that would arise from e.g. intraday optimisation, which is not currently modelled.
  - **Price (delta) correction.** To account for price variability that is not fully represented in the hourly fundamental simulations, TenneT adds a revenue uplift for batteries of 50 €/MWh (implemented via VD4 'Optionality on Shaping'). This value was derived in the MLZ 2024 by comparing the median of the maximum within-day hourly price spreads in the raw simulations (~50 €/MWh) with the corresponding value observed in the 2023 day-ahead market (~100 €/MWh).

TenneT identified the need to link these elements more closely to future market fundamentals – and this is the basis of this project.

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Comparison of TenneT & KYOS approaches

- TenneT's current EVA modelling approach and KYOS' approach were compared by performing:
  - A **qualitative** comparison of the (a) different scenario and input data assumptions, and (b) modelling steps and workflows.
  - A **quantitative** comparison of TenneT and KYOS's hourly day-ahead simulations by benchmarking for a single scenario with largely aligned scenario input data assumptions, based on the 'Basis' scenario for target year 2030 from TenneT's MLZ 2025. For simplicity, a single climate scenario (WS10) representing a typical average year for BESS revenues, and single forced outage scenario were used. Refer to Annex 1 to see the fundamental input assumptions.

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Comparison of TenneT & KYOS approaches - Qualitative

We identified the main commonalities and differences between TenneT and KYOS modeling approach

- Despite differences in modelling methodologies, both TenneT and KYOS share several foundational elements in their price simulation frameworks and BESS valuation methodologies. In fact, both approaches begin with a set of fundamental assumptions that define the scenario view of the future power system (fuel and carbon prices, installed capacities, load profiles, etc…).
- Both methodologies incorporate (Monte Carlo) day-ahead price simulations to represent uncertainty in future market conditions. In TenneT's approach the nature of these market uncertainties are driven specifically by climate variability and plant's forced outage patterns. KYOS price simulations are not based on fundamental market factors though, but on statistical methods combining e.g. multi-factor price modeling, stochastic volatility (GARCH), jumps and regime switches. For more details on the KySim model, refer to Annex 2.
- For BESS revenue assessments, KYOS uses a full-optimization software (KyBattery). The model shows what value can be made in the day-ahead, intraday, balancing and ancillary service markets and support revenue stacking strategies, respecting the complex timing and operational (state-of-charge) requirements to generate these different income streams. For more details on KyBattery, refer to Annex 3. In TenneT's case, the estimation of these value drivers are done outside of the main market model.

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Comparison of TenneT & KYOS approaches - Qualitative

TenneT

Scenario 'Base case' assumptions

- Fuel & carbon prices
- Installed capacities
- Load profiles *(climate-year dependent)*
- Etc.

→

Hourly day-ahead (DA) dispatch simulations (Monte Carlo)

PLEXOS

- Simulation of hourly dispatch and market prices for hundreds of possible realizations of the future 'Base Case' scenario based on two main uncertainties (a) different climate scenarios and (b) forced outage patterns
- Assumed to represent combined intrinsic value from forward/futures (VD1) and day-ahead (VD3) markets

→

Estimation of other value drivers (revenues)

Python/Excel

- VD2 and VD4 (optionality in forwards and intraday timeframe)
- VD5 – Ancillary services
- VD6 – Other non-electricity revenues (e.g. heat, steam, GoO)

KYOS

Scenario 'Base case' assumptions

- Fuel & carbon prices
- Installed capacities
- Load
- Etc.

→

Fundamental hourly price forward curve

KyPF

- One single deterministic hourly price forward curve (HPFC), reflecting expected future power prices based on generation costs

→

Monte Carlo simulations of forward and day-ahead prices

KySim

- Based on the fundamental HPFC from KyPF, KySim generates a set of (arbitrage free) hourly prices and dispatch results reflecting volatility of real markets, using various techniques (e.g. GARCH)
- Calibration steps for day-ahead price shapes, based on either (a) historical volatility, or (b) fundamentals

→

Simulations of intraday and imbalance prices

KySim

- Simulation (synthesis) of (a) intraday and (b) imbalance prices by sampling from historical market spreads, and applying to day-ahead prices from KySim
- Arbitrage free vs day-ahead

→

Optimization of the battery wholesale trading decisions

KyBattery

- Optimal trading strategy of battery within the intraday market (per 15 min period)

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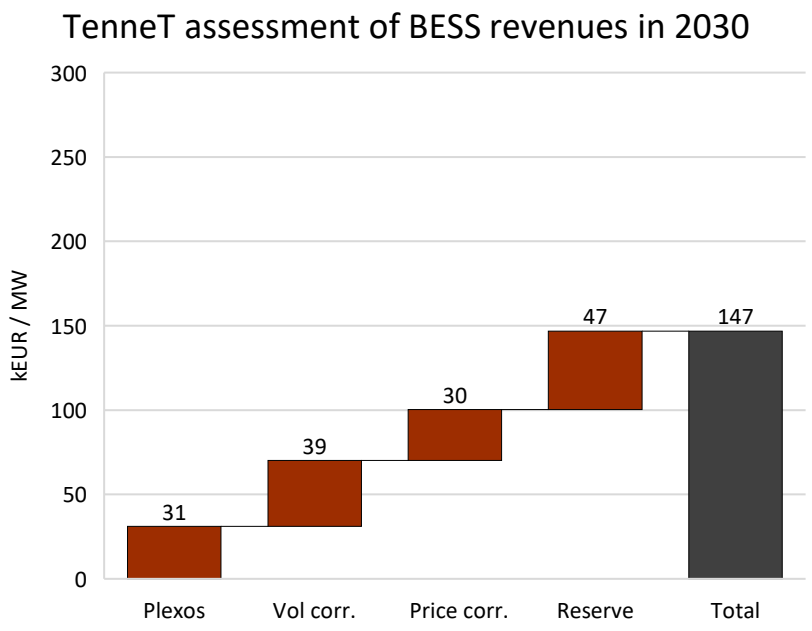
Comparison of TenneT & KYOS approaches – Quantitative comparison

TenneT assessment of BESS revenues (MLZ 2025)

- TenneT's results for the Base Case 2030 scenario are shown on the right, showing 'raw' BESS revenues from the PLEXOS hourly dispatch model at roughly 31 kEUR/MW/y.
- On top of this, TenneT applied two correction factors to better approximate BESS revenues that are not yet fully captured in the PLEXOS modelling setup (refer to slide 16). On the one hand, a volume correction factor is applied to scale BESS dispatch to around 511 cycles per year. On the other hand, a price correction factor is applied to account for the higher price volatility observed in the market.
- TenneT calculated the reserve revenues based on assumptions about the reserve volumes required by the system, the future reserve prices and by splitting the total cost for the TSO to different power plant technologies.

BESS specs: 2h duration, 90% round-trip efficiency



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Comparison of TenneT & KYOS approaches – Quantitative comparison

KYOS assessment of BESS revenues for different trading strategies

- KYOS' results on the right are derived from TenneT's hourly DA curve combined with our KySim methodology, which generate 100 Monte Carlo simulations of day-ahead, intraday and imbalance prices, calibrated on historical volatility.
- KYOS results indicate a sharp increase in value when moving from "DA only" to "ID only", driven higher price volatility and 15-min trading products. The strategies with balancing services do complement the intraday value considerably.

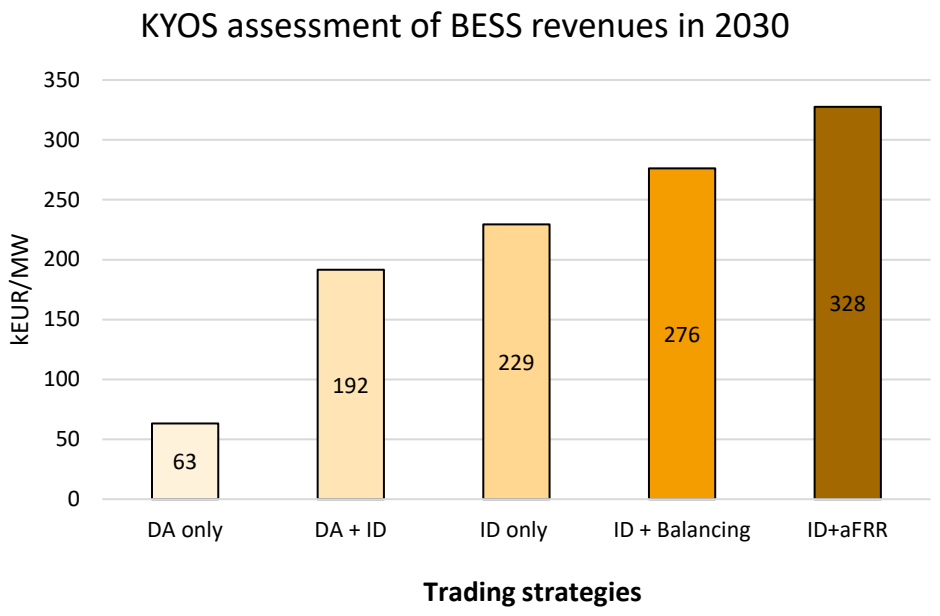
Notes:

DA + ID: Refer to a combined strategy where day-ahead trades are performed first, followed by intraday trades, if the price signal is promising enough. This might mean a partial or full reversal of a day-ahead purchase and/or an actual discharge, depending on what the nominated day-ahead position is for the same delivery period.

ID + Balancing: Strategy which combines intraday and passive imbalance trading. Intraday trades lead imbalance trades by 1 hour. The imbalance trader is not allowed to take actions that may cause nominated positions to exceed the battery capacity bounds. Imbalance volume limited to 30% of the BESS capacity.

ID + aFRR: Include revenue from aFRR upward and downward capacity as well as aFRR energy bids. aFRR capacity reservations are decided on at the day-ahead optimization stage by contrasting the expected. The rate of aFRR energy activations rate is set to 30%.

BESS specs: 2h duration, 90% round-trip efficiency, 511 cycle limits per year.



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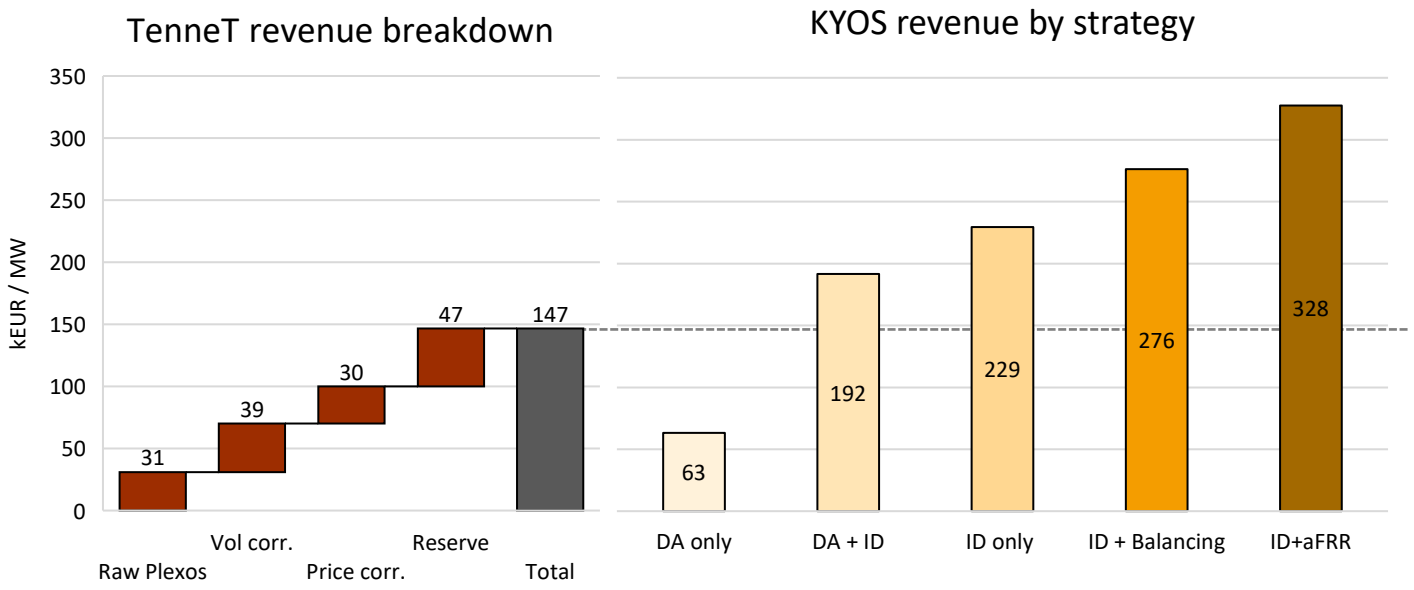
Comparison of TenneT & KYOS approaches

KYOS methodology leads to a higher BESS revenues than TenneT approach, despite its value correction factors

- We compared the outcomes from the MLZ 2025 analysis with those derived from using TenneT's day-ahead prices combined with the KYOS method for modeling BESS.
- The notable revenue discrepancy between the KYOS and TenneT methods is primarily due to intraday revenues being a key factor in the KYOS approach, whereas the "option value" in the TenneT method accounts for only a small share of total revenues.
- The higher day-ahead revenues observed in the KYOS method stem from the broader price distribution generated by our ex-post Monte Carlo simulation technique.

Comparison of 2030 BESS Revenues (Base Case Scenario, Average CY)

2h duration, 90% round-trip efficiency, 511 cycle limits.



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Comparison of TenneT & KYOS approaches - Conclusions

The focus of the process improvement should be on modelling intraday values

Based on the review of TenneT's existing approach, KYOS proposes several recommendations:

- Changes to the way TenneT performs its hourly day-ahead simulations
  - Account for additional fuel and carbon price uncertainty (e.g. additional sensitivities of input fuel and carbon price levels)
  - Explicitly model negative prices to better capture the bidding behaviour of subsidised renewables and must-run thermal plants (where feasible)
  - Include dynamic constraints of thermal power plants (minimum on/off times, minimum stable generation levels, etc…) to improve hourly price shapes (where feasible)
- An extension and refinement of TenneT's current EVA modelling framework with:
  - An additional step to optimise BESS revenues against a set of synthetic ID prices, better reflecting real-world price and volatility short-term trading (15-min granularity). In fact, the review analysis pointed to insufficient representation of intraday markets as the main cause of overall undervaluation.
  - More consistency in the way ancillary services (esp. balancing capacity) are modelled and revenues accounted for. Especially, overall value of ancillary services should reflect attainable wholesale prices due to arbitrage possibilities.

- These are outlined in more detail in the next chapter.

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Description of the proposed solution

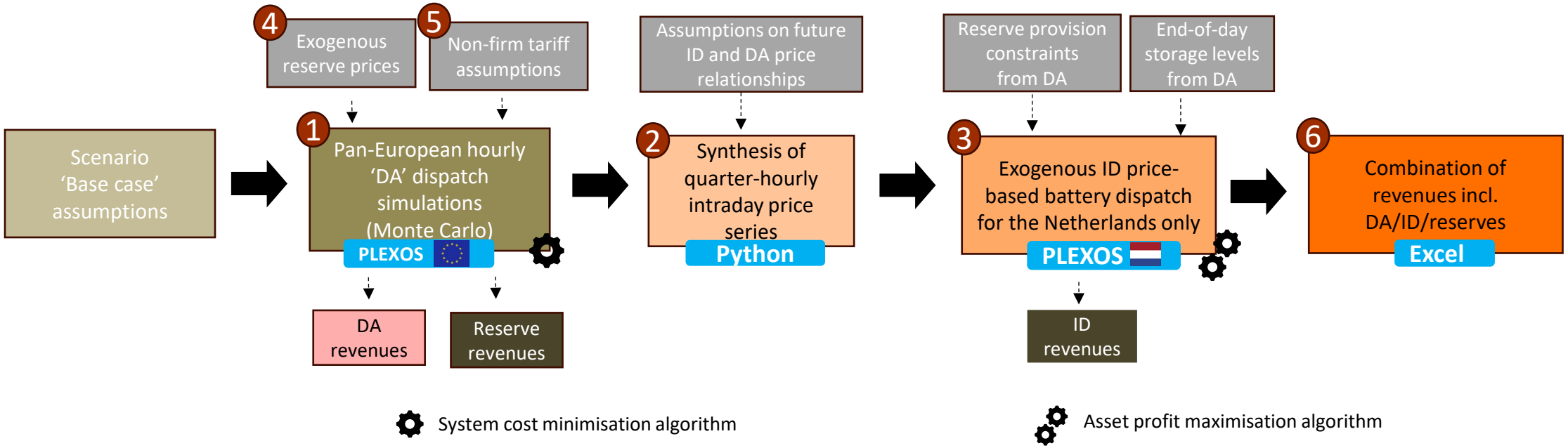
The main elements of the proposed solutions are as follows:

- Start from the fundamental scenario: The workflow begins by defining Base-case assumptions about the electricity system. Key modelling inputs such as demand, generation availability, and fuel/CO_2 price assumptions. These assumptions form the foundation for running all subsequent simulations.
- Include BESS volume restrictions for assets in the Netherlands based on assumptions for time dependent transport rights (TDTR) and Time-of-Use agreements with the TSO.
- The PLEXOS model co-optimize economic dispatch and balancing capacity reserves provision. The resulting reserves provision revenues are calculated based on exogeneous reserves price assumptions.
- Using PLEXOS, a large number of hourly Pan-European DA dispatch simulations are calculated for different climate years and random outage patterns, capturing cross-border interactions, system constraints and merit-order effects. This produces DA price and dispatch outcomes for all hours from which BESS revenues are calculated for the DA market.
- Create synthetic quarter-hourly ID price series via Python, using the simulated day-ahead results and assumptions on the DA–ID price relationships.
- In PLEXOS, a subsequent step in the approach is an ID price-based battery dispatch for the Netherlands, where volume constraints on the BESS are applied to reflect the actions taken a-priori in the DA and the balancing reserve markets.
- Finally, the method involves a weighting of the different revenue streams to form the total value of the BESS.

Overview of proposed solution approach for BESS revenues

Leveraging PLEXOS model for BESS revenue evaluations:

- The overall proposed solution ensures that it covers the value driver approach of TenneT's current methodology. It includes the BESS intrinsic value which can be derived from the electricity day-ahead market, as well as the additional value which can be derived from the intraday market and from the ancillary service markets. In the following slides, we present in more details each of the steps of the workflow below and their underlying assumptions.



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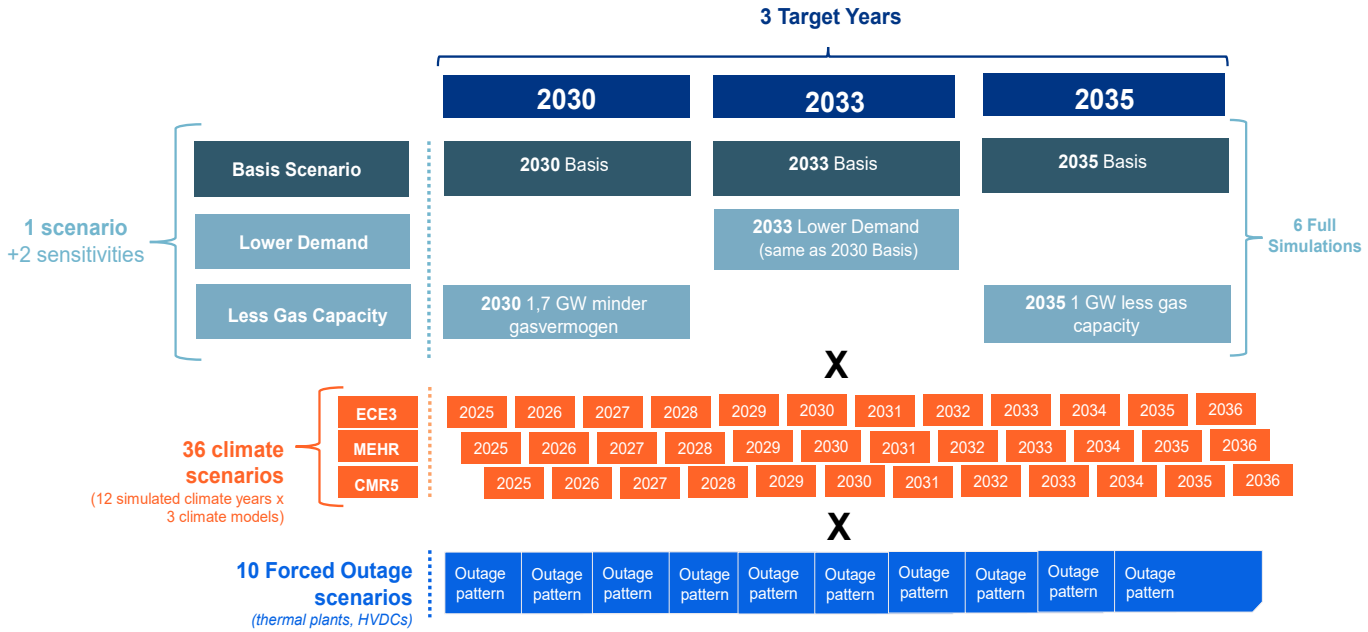
Step 1 : Recommendation on day-ahead market modelling

KYOS recommends adding sensitivities to capture a wider range of price outcomes, as well as to evaluate the impact of key assumptions and modelling choices.

- Apply the same day-ahead price modeling approach as the one currently established with a large number of climate years and forced outage patterns.
- Suggest to enhance day-ahead price modeling in future versions by:
  - Incorporating the modeling of negative prices when the model permits some level of RES curtailment.
  - Including dynamic constraints of thermal power plants
  - Including a sensitivity on fuel and carbon price levels
  - Explore a scenario with increased electrification demand.



Footnote:
Among the proposed improvements, emphasis should be placed on sensitivities to fuel and carbon prices, given their stronger influence on day-ahead price formation. The treatment of negative prices and dynamic constraints for thermal units are viewed as lower-priority refinements.

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Step 2 : Motivation for intraday modelling

An additional step to DA modeling is necessary to capture intraday market volatility

- Although the DA simulations can be conducted using different weather years and outage scenarios, PLEXOS incorporates some level of foresight regarding plant availability, so even if an outage is included, it factors into the dispatch decisions of other plants. In reality, market prices would differ between situations where a future outage is known in advance and those where it occurs suddenly and unexpectedly.
- The same reasoning apply to renewable power production forecast errors as well as demand forecast errors. Using several predefined weather patterns does not account for unexpected system overload (or underload) situations, and so the variability produced by PLEXOS simulations is not capturing intraday market price volatility.
- Consequently, an additional step is necessary to capture those pricing dynamics and other competitive factors influencing short-term residual markets. The method presented here is based on sampling historical data to introduce this missing variability. This approach is presented in the next slides.

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Step 2 : Synthesis of intraday price series

Description of the method applied to synthesize ID price series

- The methodology to synthesize intraday price series is based on a sampling technique: the method selects a historical day from a predefined period and takes the difference between intraday prices and day-ahead prices (96 data points). The method adds this absolute ID – DA spread to the future day of the day-ahead hourly price simulations and repeats the operation until the end of the curve.
- This sampling method ensures that the historical ID – DA spread come from the same weekday type as the future day. It also maintains the seasonality because the spread is taken from the same month. It ensures that the synthetic intraday price series is arbitrage free to the day-ahead price simulation. This means that for each simulation path, the ID price is equals to the DA price on average per year.
- The sampling is absolute to the day-ahead price level. Analysis of historical data has not demonstrated a strong relationship between the day-ahead price levels and the differences between DA and ID prices.
- **Throughout the report, the ID price refers to ID1 index price from the intraday continuous market. This is the average weighted price of all trades executed in the last hour before delivery. This index is used because the last hour is the most liquid (when most of the traded volumes occur).**
- The ID price series for 2030 have been validated against historical prices with different statical measurements (refer to Annex 5).

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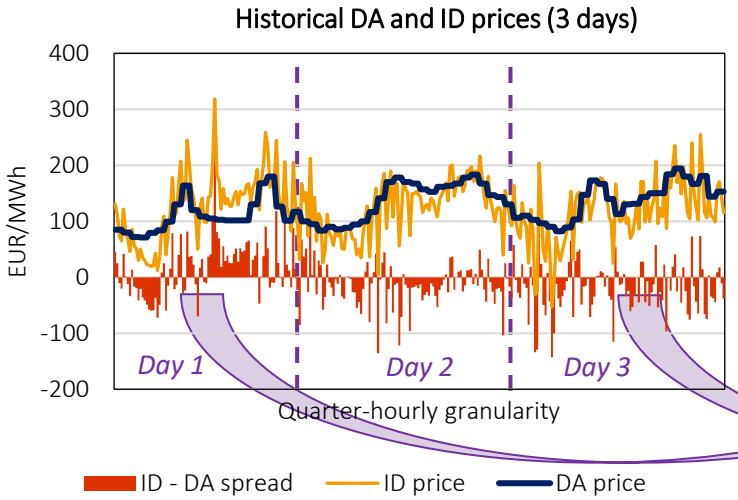
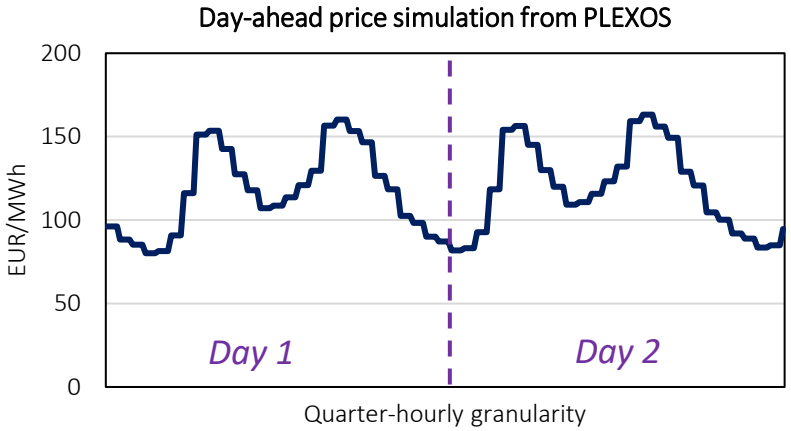
Step 2 : Synthesis of intraday price series

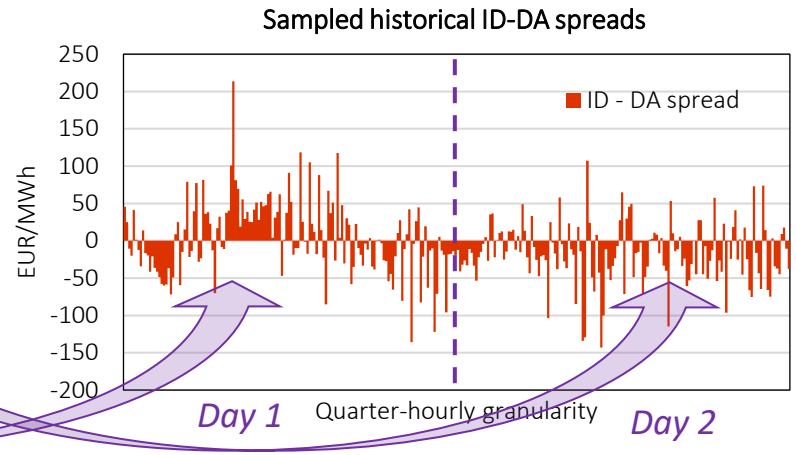
Illustration of the Python-based synthesis of ID price series

History

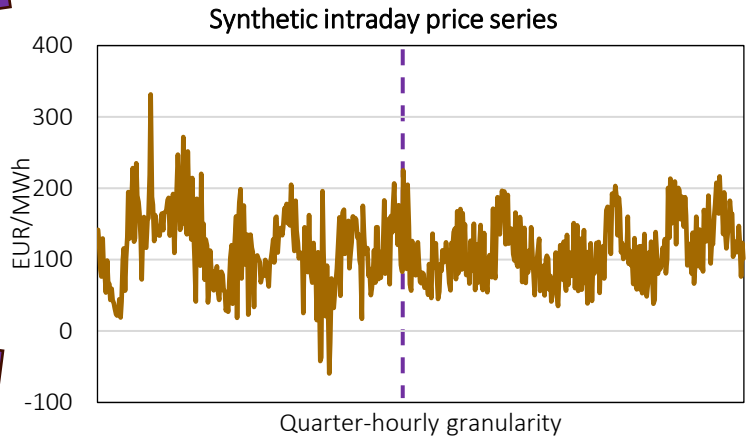
Future







ID price = Simulated DA price + sampled historical ID-DA spread



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Step 2: Adjust future ID spreads based on modelled day-ahead changes

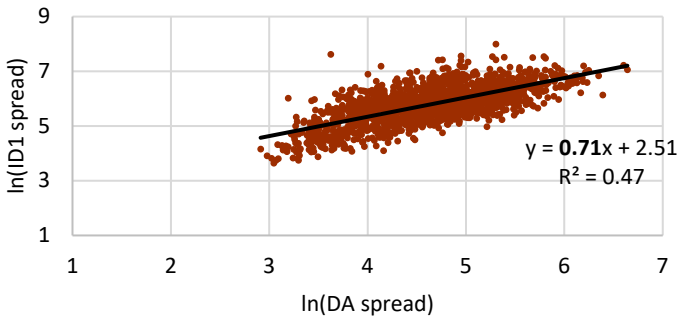
Day-ahead and intraday price spreads are correlated. The same drivers are underlying but intraday spreads are higher and more volatile due to the residual/balancing nature of the intraday market. Our approach takes this historical relationship into account.

- Day-ahead daily price spreads are calculated as the difference between the hourly maximum price minus the hourly minimum price in a day; the calculation of Intraday daily spreads follows the same logic but the highest and lowest prices have quarter-hourly granularity.
- **Intraday spreads are on average ~ 2-3x higher** than day-ahead spreads. At the time of this analysis, there isn't enough data to measure the impact of the shift to 15-min DA market.
- **Seasonality and patterns are similar** for both day-ahead and intraday daily price spreads.
- **The relationship is proportional** – a 1% change in day-ahead spreads leads to c. 0.71% change in intraday spreads (see linear equation from scatter plot).
- **The relationship is seasonal**, the difference between intraday and day-ahead spreads is larger in Q2+Q3 than Q1 and Q4.
- **The relationship is noisy**, the observed intraday spread level might be explained by the corresponding day-ahead spread only to a certain extend. This is logical because intraday spreads are frequently driven by unexpected events – e.g., RES forecast errors.

Day-Ahead and Intraday Spreads for the Netherlands*

	DA Spread [EUR/MWh	ID Spread [EUR/MWh
2021	83	194
2022	205	582
2023	108	417
2024	113	430
2025*	139	407
Avg	129	406

DA vs ID1 spread in EUR/MWh (natural logarithm)



- **Price Formation Dynamics** – Intraday prices are more exposed to short-term shocks, such as unexpected plant outages or demand surges, whereas day-ahead prices are set based on forecasts that smooth out extreme fluctuations.
- **Market Liquidity Differences** – Day-ahead markets generally have higher liquidity, and so smoother price formation.
- **Balancing Needs** – Market participants use intraday trading to correct imbalances caused by forecasting errors from day-ahead decisions or unexpected demand changes. This reactive trading can lead to price spikes as companies rush to secure electricity on short notice before delivery.
- **Regulatory and Market Design** –There are penalties for imbalances, which can drive aggressive trading in intraday markets to avoid financial losses.

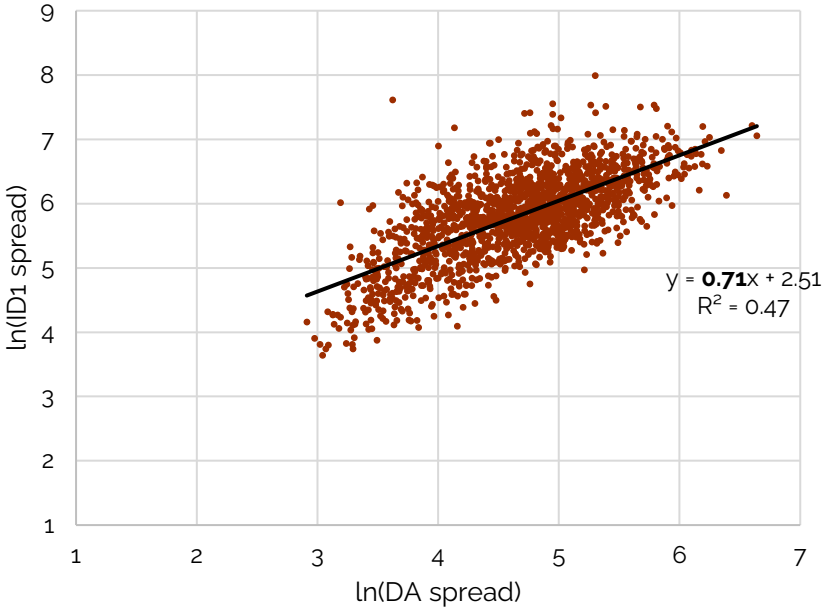
*Dataset from 01-01-2021 until 30-06-2025

Step 2: Adjust future ID spreads based on modelled day-ahead changes

Modelled day-ahead spreads provide a scenario view of the future DA-ID relationship

- Based on changes in the day-ahead spreads observed in PLEXOS we can re-scale the difference between day-ahead and intraday prices according to the historical spread relationship.
- For instance, the effect of increasing battery penetration on intraday prices is implicitly captured in the DA simulation, where the DA spread would expect to narrow. Taken in combination with the rescale factor (where 1% change in DA spread corresponds to 0.71% change in ID spreads) of historical sampling, it would also result in narrowing intraday spreads.
- This approach assumes that the historical relationship is preserved, and in particular intraday markets will be more volatile than day-ahead markets. Developments that may impact the relationship are regulatory changes (e.g., introduction of new trading products, changes in bidding rules, etc…) and structural changes (e.g. fast roll-out of flexible assets). Therefore we recommend to monitor the development of this relationship on an annual basis and update the scaling factor when necessary.

DA vs ID1 spread (natural logarithm)



*Dataset from 01-01-2021 until 30-06-2025. For each day, the natural logarithm of the DA spread is plotted against the natural logarithm of the ID1 spread.

The slope shows how sensitive intraday prices are to changes in day-ahead prices, in percentage terms. ID prices spreads react proportionally less to DA price spreads.

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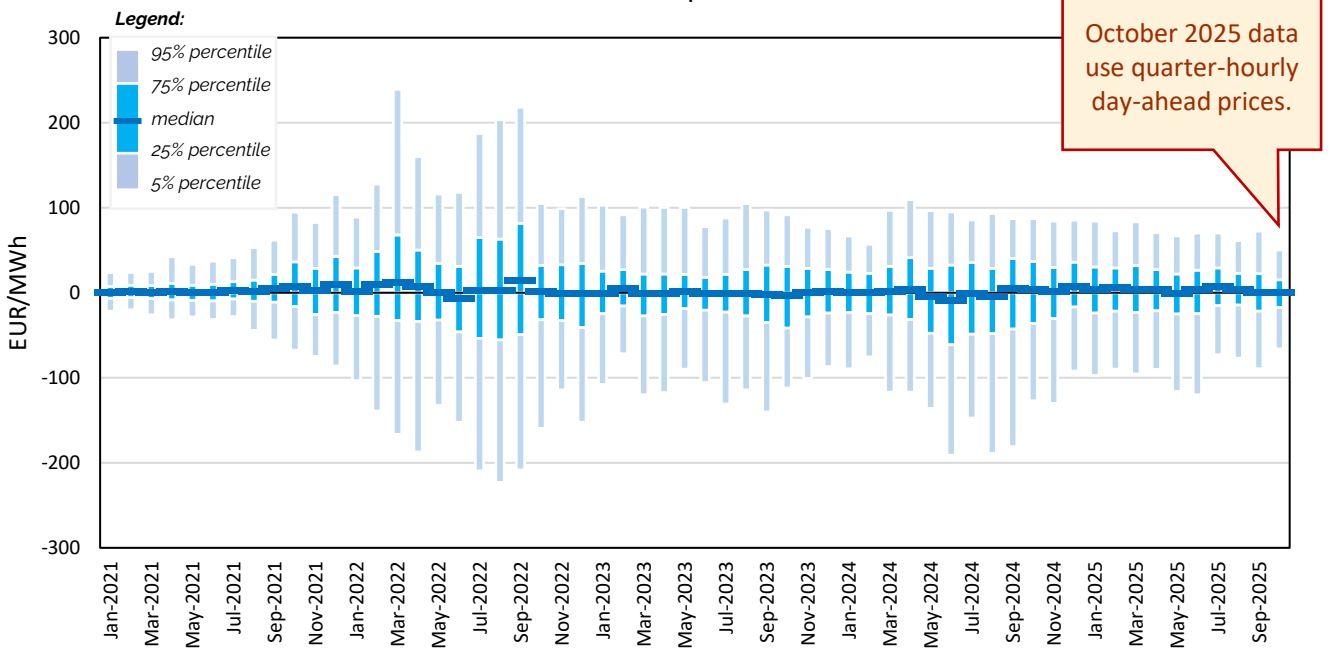
Step 2: Dealing with the future relationship between ID and DA prices

We recommend to perform sensitivities on intraday price series based on different sampling periods.

- For sensitivity analysis purposes, we recommend to sample different periods from the past to capture periods of laxness in the short-term spot markets (e.g. 2021) and tightness (e.g. 2022).
- The recent shift to 15-minute auctions in the day-ahead market may result in persistently smaller price differences between the two markets. There was insufficient data available during the project to do a full analysis, we therefore recommend to monitor this development on an annual basis. In addition, attention should be paid to the calculation granularity. Since TenneT does not plan to switch to quarter-hourly DA modeling (at least in the near term), it is appropriate to first aggregate historical DA prices to an hourly level and then perform the ID–DA sampling. We recommend to test both measures.

Note: If historical data is not sufficiently available for sampling in a year far in the past, another option would be to adjust the scaling factor. E.g. entering a rescale factor of 0.5 means the sampled DA-ID spread will be reduce by 50%, which as result means the simulated absolute ID spreads will be reduced by 50%.

Difference between DA and ID1 prices in the Netherlands



The data in the graph have been calculated as follows:

- The difference (spread) is taken between Intraday (quarter-hourly ID1) and the hourly Day-Ahead prices, per 15 minutes.
- For each month, this leads to around 4 x 24 x 30 data points.
- The statistics in the graph are the percentiles of these data points, calculated per month. For example, the 5% percentile (P5) is at a level where 5% of the spreads fall below, while the median is the middle point of the sorted spreads

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Step 2 : Synthesis of intraday price series

We developed a python script to generate intraday price series. Its principles and assumptions are outlined below.

Sampling window:

- The script samples one historical day based on the same month and day type (weekend or weekday) of the forward day. We recommend using a window of **the past 24 to 36 months** for sampling day-ahead and intraday price differences. This selection ensures sufficient variability in the dataset while excluding older periods that no longer reflect current market conditions.

```
# Sampling window (in Amsterdam time)
SAMPLING_START = "2023-01-01 00:00"
SAMPLING_END   = "2025-10-01 00:00"
```

Rescale factor:

- Since we use the absolute historical differences between DA and ID prices, a change in the DA spreads observed in the PLEXOS modelling does not automatically translate into a corresponding change in ID spreads.
- To address this, we rescale the DA–ID relationship so that changes in the projected DA spreads can be consistently reflected in the future ID spreads. This rescaling is based on the historical correlation between the two markets: a 1% change in the DA spread corresponds to a 0.71% change in the ID spread.
- We need a reference DA spread to compare against the 2030 projection, ideally from a PLEXOS back-test run for the last year. According to the KYOS modelling results, the DA spread decreases by 20% between 2026 and 2030. Accordingly, we suggest rescaling the sampled spread by -0.2 × 0.71 = **-15%.**

```
SCALING = 0.85
# Scaling factor for historic Spreads
```

Arbitrage-free:

- Based on historical data analysis, we recommend that the ID price series is considered arbitrage free to the underlying DA price simulations on a **yearly basis**.

```
# Granularity for arbitrage free adjustment
GRANULARITY = "year"   # "day" | "month" | "year"
```

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Step 3: Model parametrization for intraday price-based battery dispatch

This step involves a PLEXOS model run that applies an exogenous ID price-based BESS dispatch for the Netherlands only. The assumptions underlying this model run are outlined below:

Cycle constraints:

- We recommend to set BESS cycle constraints between **510 and 730** cycles per year for the intraday optimization (otherwise it models too many cycles). This range of cycles per year is based on typical warranty conditions provided by BESS manufacturers for systems with a 15-year technical lifetime.

End-of-day BESS set points from day-ahead model run:

- We recommend enforcing the end-of-day (EoD) state-of-charge level produced by the DA model as a constraint in the ID optimization model run. To some extent, this preserves consistency between DA market commitments and ID trading by ensuring the BESS arrives at the DA planned energy level at the end of the trading day. This is important to avoid profitable ID cycles that would violate prior DA commitments.

Accounting for reserve provision in intraday trades:

- We recommend to pass on the hourly reserve provision allocated to batteries from the DA model into the ID model. This ensures that for the time periods when BESS are awarded to provide reserves, wholesale trading (both DA and ID) revenues can not be performed.

Bid-ask spreads:

- We recommend considering higher liquidity (or trading) costs when trading intraday (compared to day-ahead), to better reflect the actual liquidity conditions that market participants are facing. To approximate this, we suggest to implement a higher variable operations and maintenance (VOM) costs for BESS assets in the intraday model run. This value should be based on a calculation of ID Bid-Ask spreads observed in the same historical period used in the ID sampling. The next slide presents an example.

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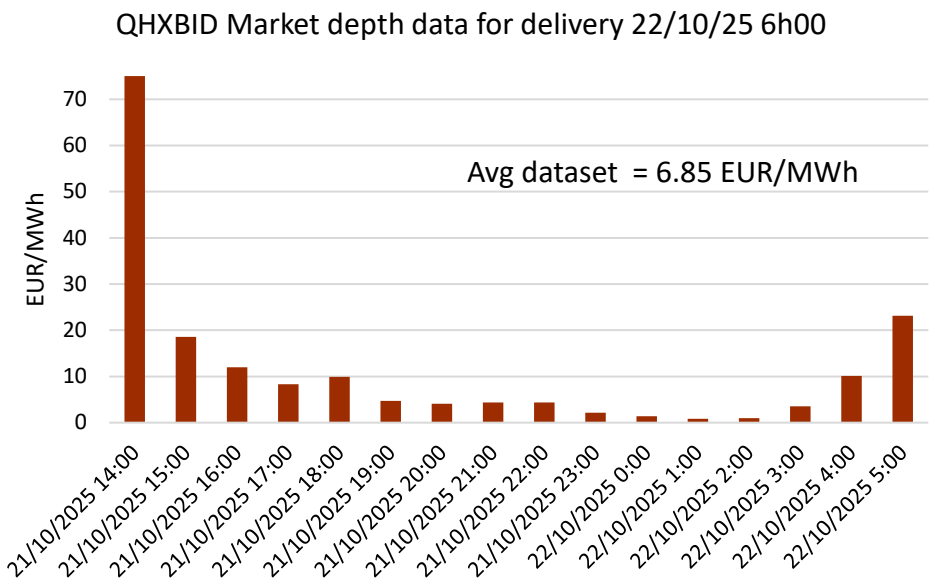
Step 3 : Motivation for liquidity costs in the intraday market

Use real order book data as a basis for the VOM cost

- The liquidity cost refers to the implicit cost of trading caused by limited market depth, rather than an explicit fee. It mainly shows up in the price you get versus the “true” or mid-market value and is measured as a bid-ask spread. The figure on the right provides an illustration from real order book data.
- We recommend to use the volumed weighted bid-ask spread (available volume at the bid/ask) as a basis for the liquidity cost.
- **How to implement liquidity costs in PLEXOS?**
- By default, PLEXOS assumes a single ID market price for both buying and selling energy within a given period. To represent a bid–ask spread of 10 EUR/MWh for example, the purchase price is increased by +5 EUR/MWh and the sale price is decreased by -5 EUR/MWh.
- This logic is approximated and implemented by applying variable operating and maintenance (VOM) costs to the BESS, thereby introducing an effective trading cost that captures intraday market liquidity constraints. On the basis of an analysis of a sample order book data, we recommend to apply a VOM cost of 5 EUR/MWh to BESS.

“Bid-ask spread”: Min Sell price minus Max Buy price



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Step 4 : Ancillary service markets

Modeling of the ancillary service market revenues for BESS

- 1) **Reserve market representation**: Ancillary service revenues are incorporated into the modeling framework through an exogenous representation of reserve prices and an endogenous allocation of reserve capacity within the DA optimization. Specifically, exogenous reserve prices are provided for **three reserve types**
  - (a) Spin up: FCR + aFRR
  - (b) Spin down: FCR + aFRR
  - (c) Non-spin up: mFRR
- 2) **Reserve price assumptions**: Reserve prices are set exogenously based on the estimated opportunity costs of thermal generation units and BESS, and are supported by observed historical price trends (see next slide for actual price assumptions).
- 3) **Reserve allocation and BESS capacity constraints**: The DA model endogenously determines the allocation of reserve capacity to eligible thermal and BESS assets. For BESS, ancillary service provision is constrained to a maximum of **50%** of installed power capacity. This limitation ensures that sufficient power capacity remains available for state-of-charge management and energy arbitrage, thereby preventing infeasible operating schedules. On the other hand, This threshold is also based on testing PLEXOS reserve provision outcomes with and without constraints on the capacity.
- 4) **Eligibility of BESS for reserve products**: BESS are allowed to provide FCR and aFRR, both in upward and downward directions, but are excluded from mFRR provision. This modeling choice reflects the technical strengths of BESS, which are particularly well suited to fast-response services such as FCR and aFRR. In contrast, mFRR typically features lower capacity prices and smaller activation volumes, making it less attractive for BESS participation under expected market conditions.
- 5) **Treatment of imbalance energy**: Imbalance energy revenues (whether from aFRR energy bids or passive imbalance trading) are not explicitly quantified in the model. These revenue steams are characterized by smaller volumes and less predictable price formation. Excluding imbalance energy revenues therefore results in a conservative estimate of the total BESS revenues.

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Step 4 : Ancillary service markets

Principles to determine the exogenous reserve capacity prices

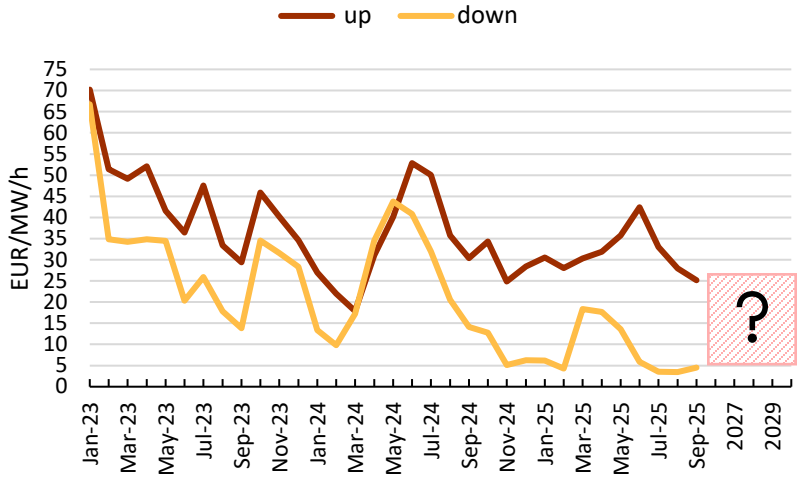
- **Recent development of reserve prices:** aFRR capacity prices in the Netherlands have declined in the last 2 years.
- **Methodology to determine an exogenous aFRR capacity price forecast:** We cross-checked both TenneT and KYOS approach
- **Historically prices mainly set by thermal power plants** and prices are determined by the foregone profits from sales in the day-ahead market.
- *Hirth & Ziegenhagen (2013): The opportunity costs of providing positive reserve, for example SC+, can be written as a function of the spot price p, the plant's variable cost c, the minimum load Pmin and the amount of control power the plant can deliver Psc:*

$$sc^{+} = \begin{cases} (p-c) & \text{if } p > c \\ -(p-c) * P_{min}/P_{SC} & \text{if } p < c \end{cases}$$

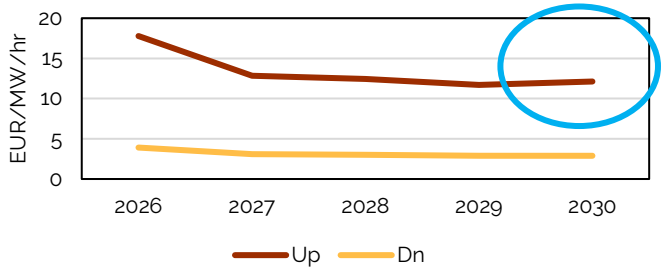
KYOS in-house approach: Similar principle but looking at market level, not plant level

1. Linear regressions between historical aFRR prices and clean spark spreads (CSS), for different gas generation levels.
2. Apply regression coefficients to forecasted CSS and gas production levels from DA model run. This determines the cost for gas plants to provide the reserve.
3. Opportunity costs for BESS is derived from the intraday trading value.
4. aFRR capacity price determined by the lowest cost offered between BESS and gas plants.

Historical aFRR capacity prices in NL



KYOS aFRR capacity price forecast



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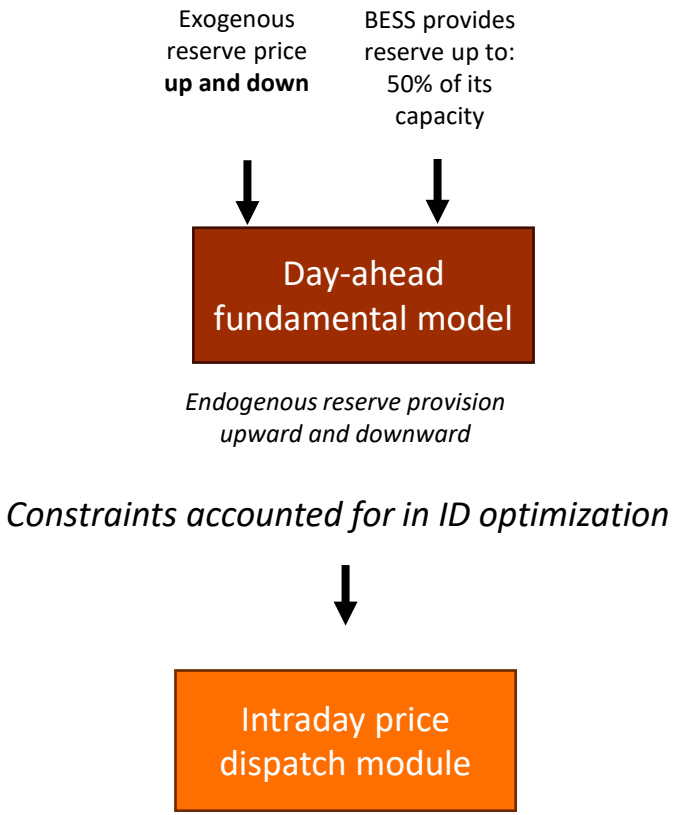
Step 4 : Model parametrization for reserve provision

Modelling of ancillary market revenues

Implementation

- **We refer to opportunity cost approach to derive the exogenous reserve price for upward & downward directions:** For the 2030 test case, we suggest using an exogenous **upward** reserve price of **12 EUR/MW** and assume a lower exogenous **downward** reserve price of ~ **5 EUR/MW**.
- **Maximum share of BESS capacity**: We recommend to limit the share of the BESS capacity able to provide capacity reserve to 50%. This value is based on testing PLEXOS reserve provision outcomes with and without constraints on the capacity.
- **Accounting for reserve provision in intraday trades**: We recommend to pass on the hourly reserve provision allocated to batteries from the DA model into the ID model. This ensures that for the time periods when BESS are awarded to provide reserves, wholesale trading (both DA and ID) revenues can not be performed. Double counting is, therefore, avoided.



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Step 5: Account for TDTR & Time-of-use tariffs (1)

How are time-of-use grid tariffs likely to affect BESS dispatch and how to account for it in PLEXOS?

- In order to lower the fixed component of their grid fees, BESS project developers are likely to conclude time dependent contracts with TenneT, for the TSO to use scarce grid space more efficiently. With a time dependent transmission right (**TDTR)** contract, TenneT can impose restrictions on the **charge and/or the discharge rate** of the BESS during certain hours of the year (up to 15% of the time). These restrictions will be communicated at least one day in advance. It therefore implies to account for it in the day-ahead modeling step.
- The regulator ACM also published the **Time-of-Use** weighting factors, which can be exploited by BESS assets to reduce the kWmax component (actual usage component) of their grid fees. The **kWmax tariff** is set by the highest charge rate in each month. The hourly charge rate is first weighted by factors as shown in the table below. **Grid fees are minimized if the consumption in the hours with the highest weighting factors is reduced to a level that makes the weighted consumption the same as in the hours with the lowest weight.**
- For example, in January the maximum discount applies if the charge rate is reduced to 70% between 09:00 and 22:00. Additionally, consumption should be reduced to 80% and 90% in 4 more hours of a normal weekday.

- These Time-of-Use patterns can be used to minimize the applicable kWmax fee for a BESS by limiting its nominal charging capacity. Two approaches are worth considering:
  - Restrict the BESS charge rate sufficiently to fully minimize the kWmax fee.
  - Or allow higher charging flexibility by not targeting the lowest weighting factor (0.6), but accepting a slightly higher kWmax fee.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Jan	0.7	0.7	0.7	0.7	0.7	0.7	0.8	0.9	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.9	0.8
Feb	0.7	0.7	0.7	0.7	0.7	0.7	0.8	0.9	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.9	0.8
Mar	0.7	0.7	0.7	0.7	0.7	0.7	0.8	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	1.0	1.0	1.0	1.0	0.9	0.8	0.8
Apr	0.7	0.7	0.6	0.6	0.6	0.6	0.7	0.8	0.8	0.7	0.6	0.6	0.6	0.6	0.6	0.6	0.7	0.8	0.8	0.8	0.8	0.8	0.8	0.8
May	0.7	0.7	0.6	0.6	0.6	0.6	0.7	0.8	0.8	0.7	0.6	0.6	0.6	0.6	0.6	0.6	0.7	0.8	0.8	0.8	0.8	0.8	0.8	0.8
Jun	0.7	0.7	0.6	0.6	0.6	0.6	0.7	0.8	0.8	0.7	0.6	0.6	0.6	0.6	0.6	0.6	0.7	0.8	0.8	0.8	0.8	0.8	0.8	0.8
Jul	0.7	0.7	0.6	0.6	0.6	0.6	0.7	0.8	0.8	0.7	0.6	0.6	0.6	0.6	0.6	0.6	0.7	0.8	0.8	0.8	0.8	0.8	0.8	0.8
Aug	0.7	0.7	0.6	0.6	0.6	0.6	0.7	0.8	0.8	0.7	0.6	0.6	0.6	0.6	0.6	0.6	0.7	0.8	0.8	0.8	0.8	0.8	0.8	0.8
Sep	0.7	0.7	0.6	0.6	0.6	0.6	0.7	0.8	0.8	0.7	0.6	0.6	0.6	0.6	0.6	0.6	0.7	0.8	0.8	0.8	0.8	0.8	0.8	0.8
Oct	0.7	0.7	0.7	0.7	0.7	0.7	0.8	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	1.0	1.0	1.0	1.0	0.9	0.8	0.8
Nov	0.7	0.7	0.7	0.7	0.7	0.7	0.8	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	1.0	1.0	1.0	1.0	0.9	0.8	0.8
Dec	0.7	0.7	0.7	0.7	0.7	0.7	0.8	0.9	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.9	0.8
W'end/holiday	0.7	0.7	0.6	0.6	0.6	0.6	0.7	0.8	0.8	0.7	0.6	0.6	0.6	0.6	0.6	0.6	0.7	0.8	0.8	0.8	0.8	0.8	0.8	0.8

ACM: Time-of-Use weighting factors

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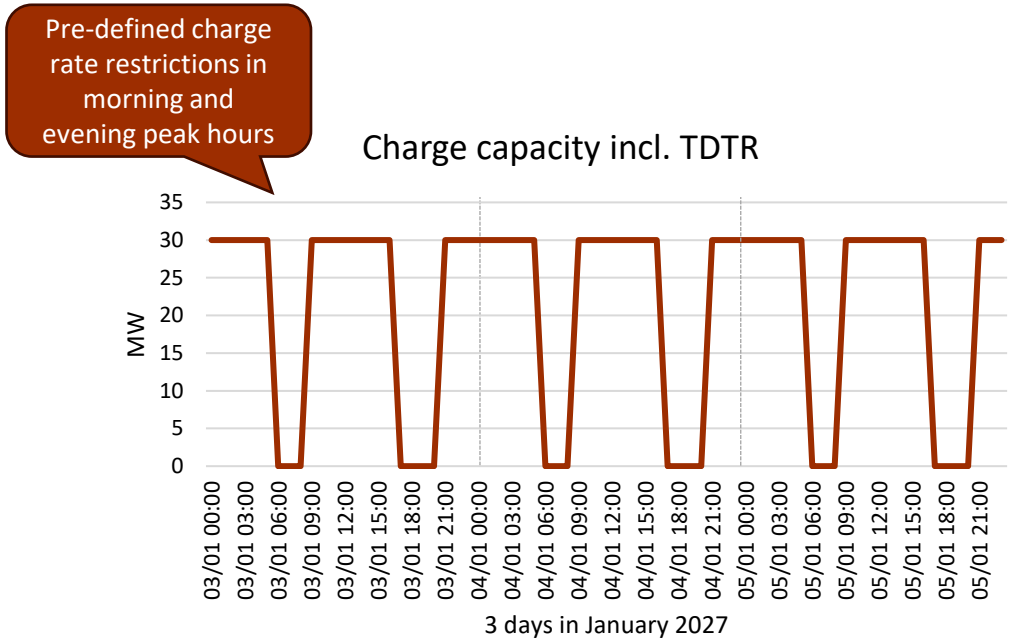
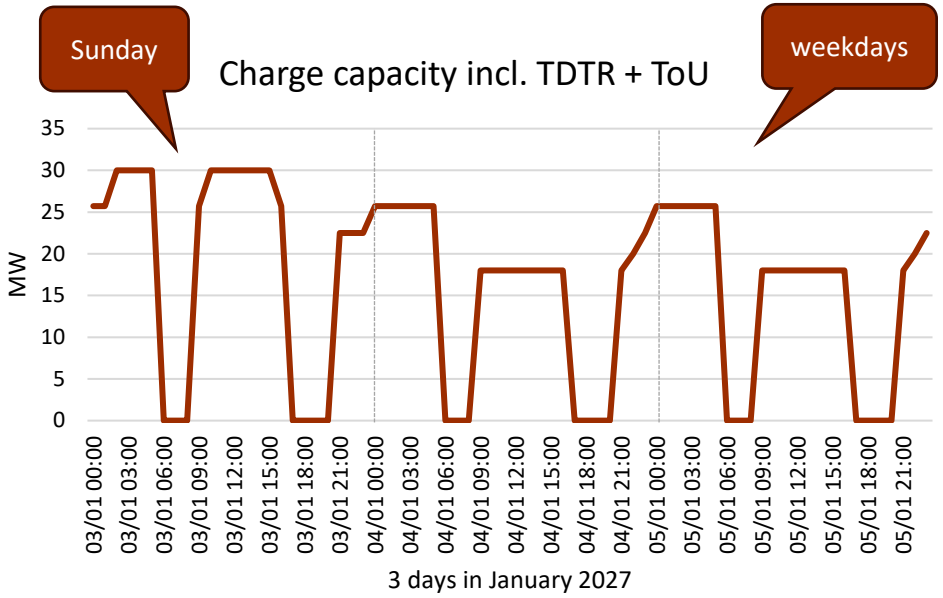
Step 5 : Account for TDTR & Time-of-use tariffs (2)

How are time-of-use grid tariffs likely to affect BESS dispatch and how to account for it in PLEXOS?

- We recommend **to specify BESS volume constraint directly into the DA optimization (and to be passed on the ID optimization as well).** This can be uploaded as an hourly timeseries.
- Below is an illustration of charge rate limitations on a 30 MW BESS, using assumptions from the matrix in the previous slide.





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Step 6 : Avoiding double-counting wholesale revenue streams

Modelling of wholesale market revenue stacking

- Modelling BESS dispatch decisions follows a sequential logic, where choices in each market depend on previous trades.
- Separately modelling DA and ID could lead to double counting unless BESS capacity is clearly divided between the markets.
- We evaluated following two alternatives:

1 Split by Current Market Volumes

- Highly transparent
- Tends to place too much emphasis on day-ahead markets while undervaluing intraday and ancillary services, making it a cautious strategy based on anticipated market behavior.
- Challenging to predict how liquidity between day-ahead and intraday markets will evolve over time.
- Offers some protection against the risk of low liquidity in the intraday market.

2 Split by Trading Preferences

- Will reflect market behaviour BUT market behaviour is expected to change as BESS grows in capacity and A/S markets are saturated.
- No or limited public indices with market segment breakdowns for the NL
- This could be mitigated by engaging with market participants in a survey – similar to how other generation assumptions are collected.

- Note that possible future development of the approach would be to have an integrated DA and ID trading mechanism, which would avoid splitting the BESS capacity and modeling these markets separately.

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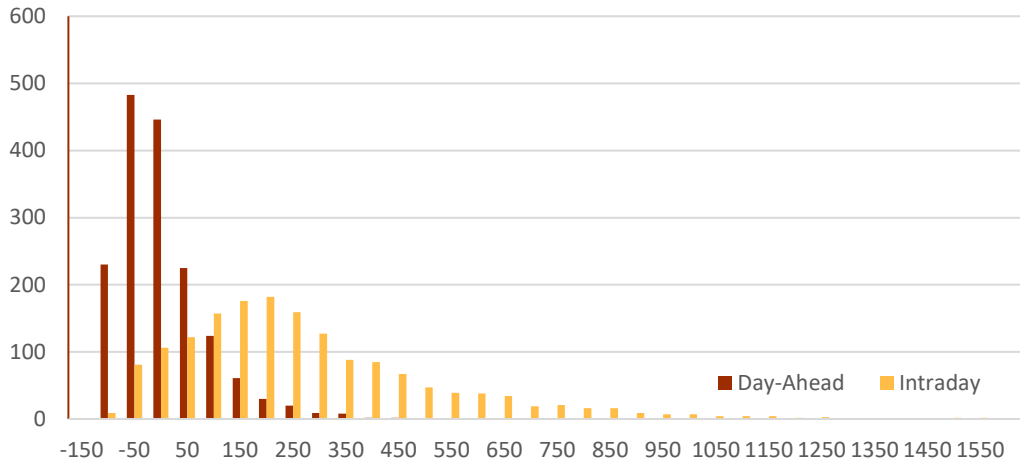
Step 6 : Avoiding double-counting wholesale revenue streams

Modelling of wholesale market revenue stacking

- Splitting trading between day-ahead and intraday markets arbitrarily is challenging because it requires using the average spreads in each market as a reference. In practice, trading decisions are more complex and harder to capture within this framework.
- Feedback from BESS developers indicates that intraday markets are more appealing due to their higher volatility.
- Average day-ahead spreads are lower than those in intraday markets, this drawback is not offset by lower variability in returns. Consequently, most portfolio strategies would tend to favor the intraday market. (See chart and academic citation on the right-hand side)
- **For wholesale trading alone, we suggest allocating approximately one-third to day-ahead and two-thirds to intraday markets, based on KYOS's modeling results, which support asset co-optimization across both markets.**
- **Nonetheless, this allocation is somewhat arbitrary; therefore, we recommend reviewing this assumption annually with market participants during TenneT's surveys.**
- **While this method might place slightly more emphasis on the day-ahead market, it also helps address the risk that intraday markets may lack sufficient liquidity to trade all BESS capacity, making it a cautious and reasonable approach. It may also compensate for the perfect foresight in the modelling.**

Distribution of market spreads (2021 - 1H2025)



"*Multi-market bidding strategies that include participation in the day-ahead market tend to perform similarly or even less favorably compared to strategies that focus solely on intraday auction trading, followed by redispatching via the rolling intrinsic approach in the continuous intraday market.*

While the results […] indicate that the standard deviation of profits is lower for day-ahead bidding relative to intraday or rolling intrinsic strategies, ***this lower variability does not reflect improved risk-adjusted performance****. Rather, it is a consequence of a reduced range of attainable profits, particularly on the upside, which is not advantageous from an economic standpoint.*"

Source: Oeltz and Pfingsten, Rolling intrinsic for battery valuation in day-ahead and intraday markets, October 2025

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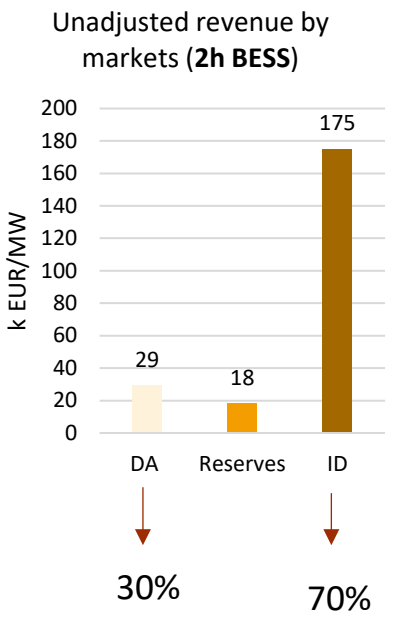
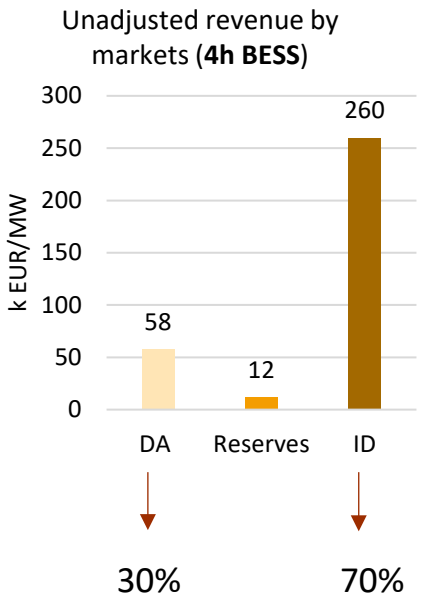
Results for individual market segments and stacked revenues

Results from a single price scenario for the year 2030 Base Case Scenario for an average climate year

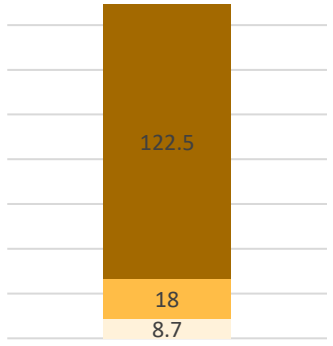
- **The results showed in the bar charts on the right are underlined by:**
- Synthetic ID price series rescaled by a factor of 0.85.
- Reserve provision allocated to BESS from the DA model output, as well as EOD battery storage level, are passed on into the ID optimization.
- Included a VOM cost of 10 EUR/MWh for intraday optimization.
- Exogeneous upward reserve prices of 12 EUR/MW/h and downward price of 5 EUR/MW/h and a limit to BESS reserve provision of 50% of the asset capacity.
- BESS are allowed to provide FCR and aFRR, both in upward and downward directions, but are excluded from mFRR provision.

- **The figures on the right show the results based the recommended split of BESS capacity allocation between the day-ahead (30%) and the intraday market (70%).**
- This weighting factor is based on KYOS's modeling results, which support asset co-optimization across both markets. It might place slightly more emphasis on the day-ahead market. It helps address the risk that intraday markets may lack sufficient liquidity to trade all BESS capacity, making it a cautious and reasonable approach.
- The revenues from reserve provision can be added to the revenue streams because the approach ensures that double-counting is avoided.

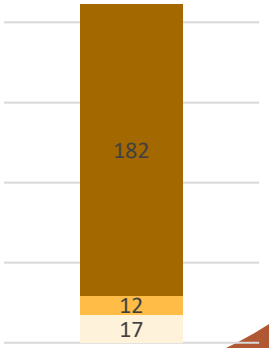




149 kEUR/MW/year



211 kEUR/MW/year



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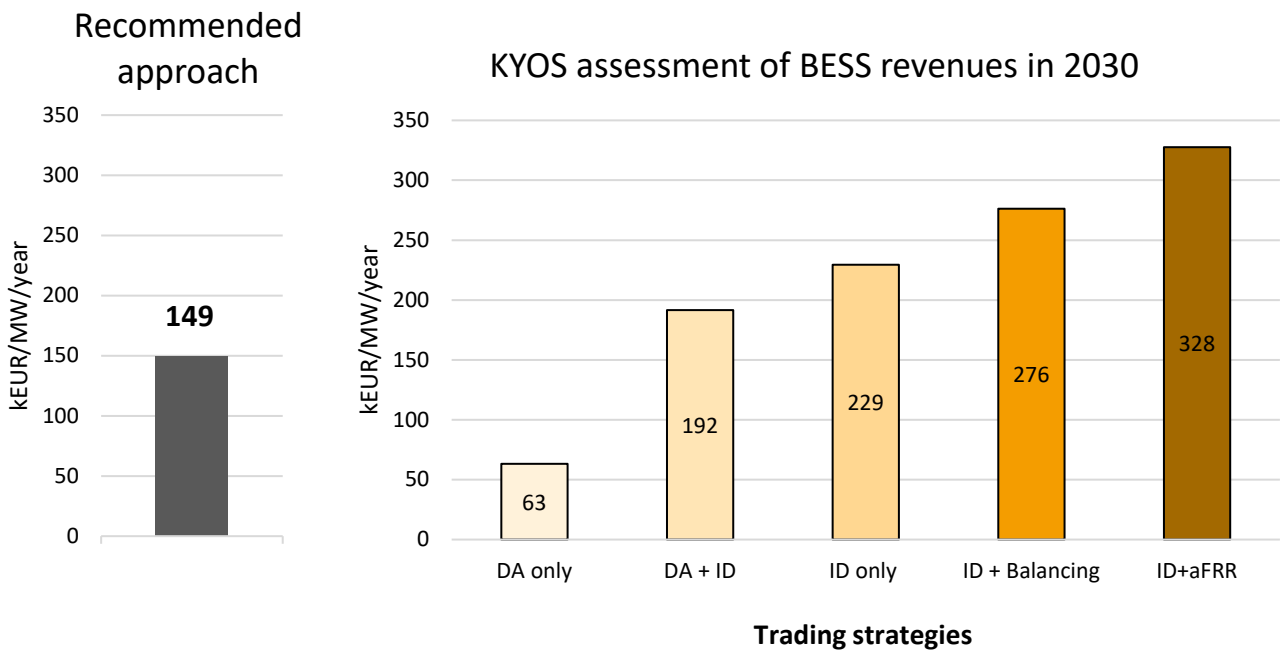
Results comparison against KYOS benchmarks

Revenues from the recommended approach against different trading strategies from KYOS KyBattery model

- The BESS revenue results of the recommended approach for the Base Case Scenario lie considerably below KYOS benchmark (-28%) for a trading strategy that combined DA and ID trading, for a similar asset.
- While the methodologies are different, this suggests that the recommended approach is conservative in the evaluation of intraday revenue especially, thanks mainly to the following measures taken in the recommended approach:
  - Rescale factor of 0.85 (which was not applied to KYOS benchmark results)
  - BESS volumes constrained by the reserve provision from the DA model, as well as EOD battery storage level, that are passed on into the ID optimization.

BESS specs: 2h duration, 90% round-trip efficiency, 511 cycle limits per year.



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Conclusions

Enhanced framework integrating key Revenue streams with market-based validation

- The project delivers a comprehensive methodology for the TenneT team to evaluate utility-scale battery revenues and business cases consistent with TenneT's MLZ Report. The methodology builds on TenneT's existing PLEXOS framework for day-ahead modeling, with supplementary elements implemented in accessible tools (python script and excel files) using a workflow that can be applied consistently across all study years.
- The approach integrates relevant revenue streams—day-ahead markets, intraday markets, balancing capacity, and ancillary services—through a structured workflow that combines insights from PLEXOS fundamental market analysis with Python-based tools for intraday price synthesis and revenue optimization.
- The Python-based sampling technique generates quarter-hourly intraday price series that maintains arbitrage-free consistency with day-ahead simulations while capturing the higher volatility characteristic of residual markets. A rescale factor based on historical correlations ensures that fundamental changes in day-ahead markets are appropriately reflected in intraday valuations, maintaining the link to market fundamentals. This proportional relationship reflects historical analysis showing intraday spreads are typically 2-3x higher than day-ahead spreads.
- Reserve provision is calculated within the PLEXOS day-ahead model, with exogenous pricing based on opportunity cost logic linked to day-ahead and intraday modeling results. These reserve commitments are passed through as constraints to the intraday optimization, preventing double allocation of capacity.
- The methodology addresses revenue stacking opportunities while recognizing participation constraints through sequential optimization logic that prevents double counting and reflects market depth.
- Ongoing engagement with market participants (particularly regarding market strategy) will help validate and refine the modeling approach over time.

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Glossary

Abbreviation	Term
BESS	Battery Energy Storage Systems
EVA	Economic Viability Assessment
DA	Day-ahead
ID	Intraday
ID1	The weighted average price of all continuous trades executed within the last trading hour
UCED	Unit Commitment and Economic Dispatch
MLZ	Security of Supply Monitor (“Monitoring Leveringszekerheid”)
V2G	Vehicle-to-Grid
VOM	Variable operations and maintenance
VD	Value Driver
TDTR	Time dependent transmission rights

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Annex 1 – Base Case scenario assumptions for 2030

Overview of power market fundamental assumptions in 2030, used for benchmarking in this study

- TenneT Base Case scenario is a top-down scenario largely in line with national and EU goals for deployment of RES and electrification to met net-zero targets by 2050.
- The assumptions for the Netherlands are based on multiple sources while the assumptions outside NL are based on the European Resource Adequacy Assessment 2024.
- While KYOS' own 'Base Case' followed the same trends, it had different fundamental assumptions and has adapted it to align with TenneT assumptions in order to perform accurate comparisons.

Parameter	Base Case assumptions** for TY 2030 (MLZ 2025)			
Unit: GW *if not specified*	NL	BE	DE	FR
Solar	54	14	97	40
Onshore wind	8	5	115	31
Offshore wind	12	6	28	4
Demand (TWh) *	140	112	516	533
Thermal capacity				
Nuclear	0.5	2	0	63
Coal & lignite	0	0	12	0
Natural Gas	14	6.5	31	7
Fuel prices				
Gas (EUR/MWh)	22.4			
Coal (USD/t)	76			
CO2 (EUR/t)	137			
H2 (EUR/t)	2'800			

* Excluding electrolyser consumption

** For selected countries. For more data, check: TenneT Dashboard

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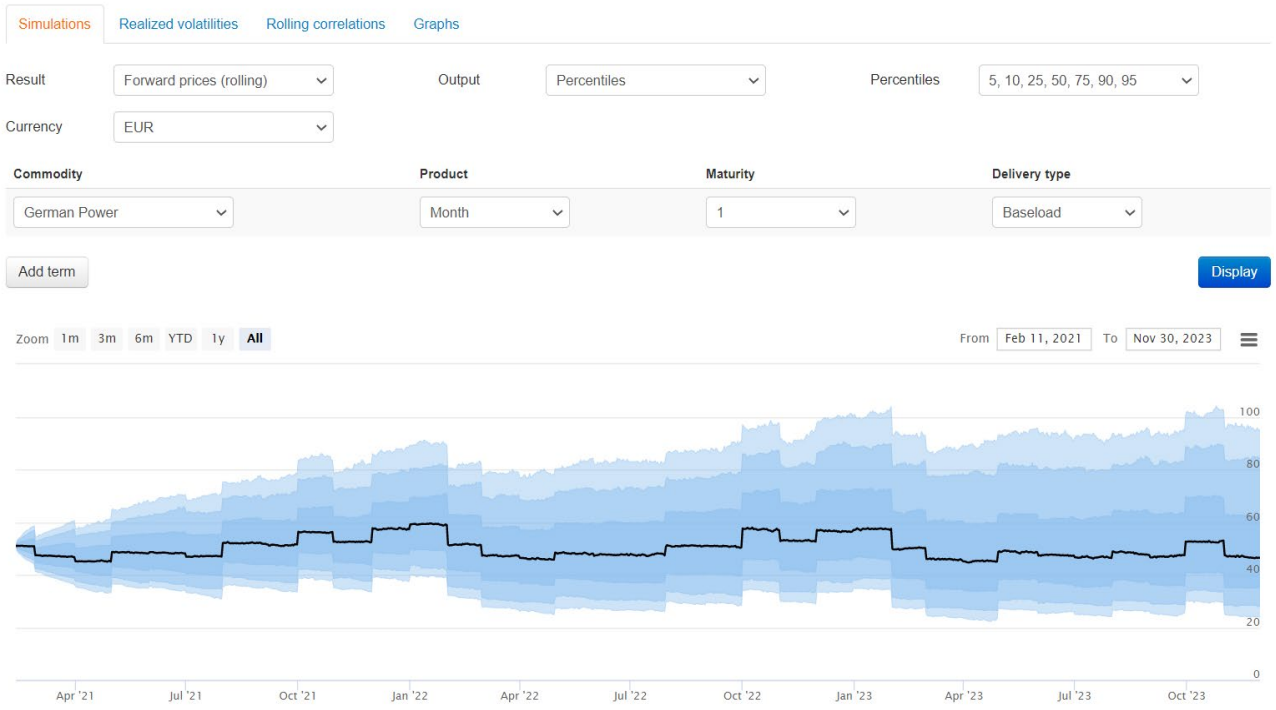
Annex 2 – KYOS Monte Carlo price simulation model “KySim”

Overview of KYOS price simulations engine “KySim”

- In order to assess price risk, it is essential to assess the price on many simulated paths. The KySim model generates such forward and spot price simulations around an hourly power price forward curve. The model has been developed and continuously improved since its first version was released in 2004. It combines a number of methodologies like multi-factor price modeling, co-integration, stochastic volatility (GARCH), jumps and regime switches. The methodologies are well founded in the academic literature, including various publications by KYOS authors.
- Mean-reversion is taken into account, meaning that shocks of the spot price persist only for a few days. Daily spot prices are mean-reverting to the front-month forward prices.
- The volatility levels are historically calibrated on market prices.
- The output of the Monte Carlo simulation model are both forward and daily/hourly spot prices.



Example: Percentiles of simulated month-ahead forward prices in the German power market

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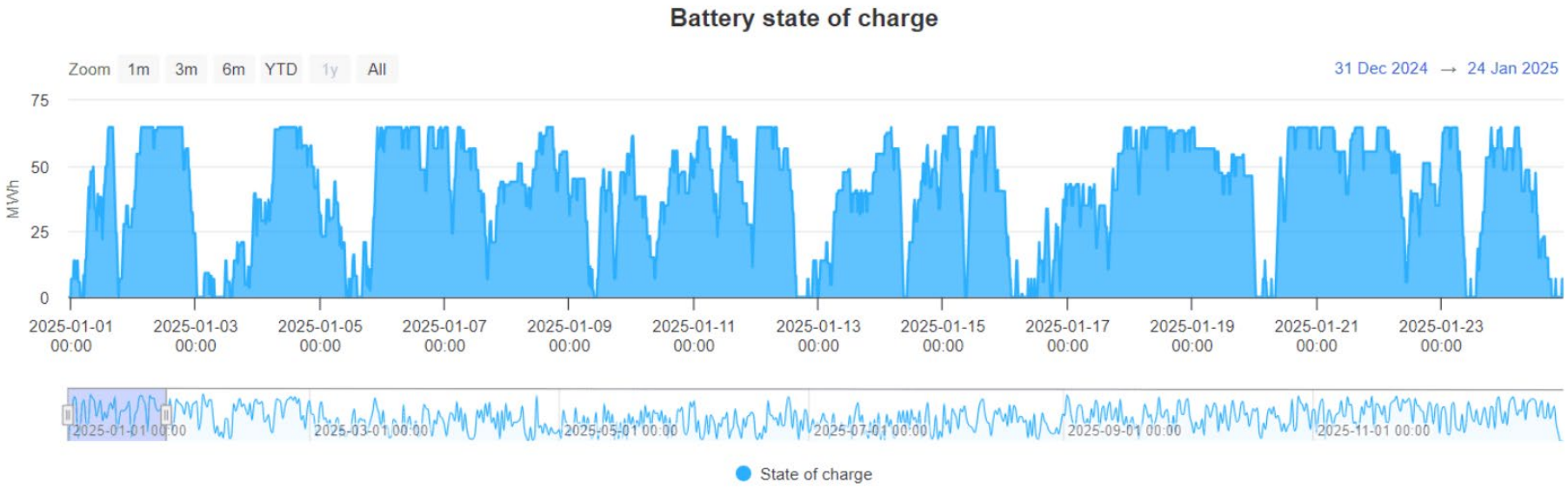
Annex 3 - KYOS Battery optimization model “KyBattery”

Main principles of the KyBattery software

- The KyBattery energy optimization software includes all common energy storage parameters: time dependent charge and discharge rates, costs and efficiencies, battery degradation, limits to the number of cycles, and reduced access to the grid.
- The model calculates the fair value of energy storage. The model calculates the intrinsic value of the battery, derived from an optimal dispatch against the hourly (or quarter-hourly) price forward curve. The extrinsic value is the additional value that can be derived on average when the battery is dispatched against the Monte-Carlo simulated prices. The optimization method used to calculate the battery dispatch on the large number of price pathways can either be Dynamic programming (DP), a method with perfect foresight of future prices, or the Least-Square Monte Carlo method (LSMC), an optimization method without perfect foresight.



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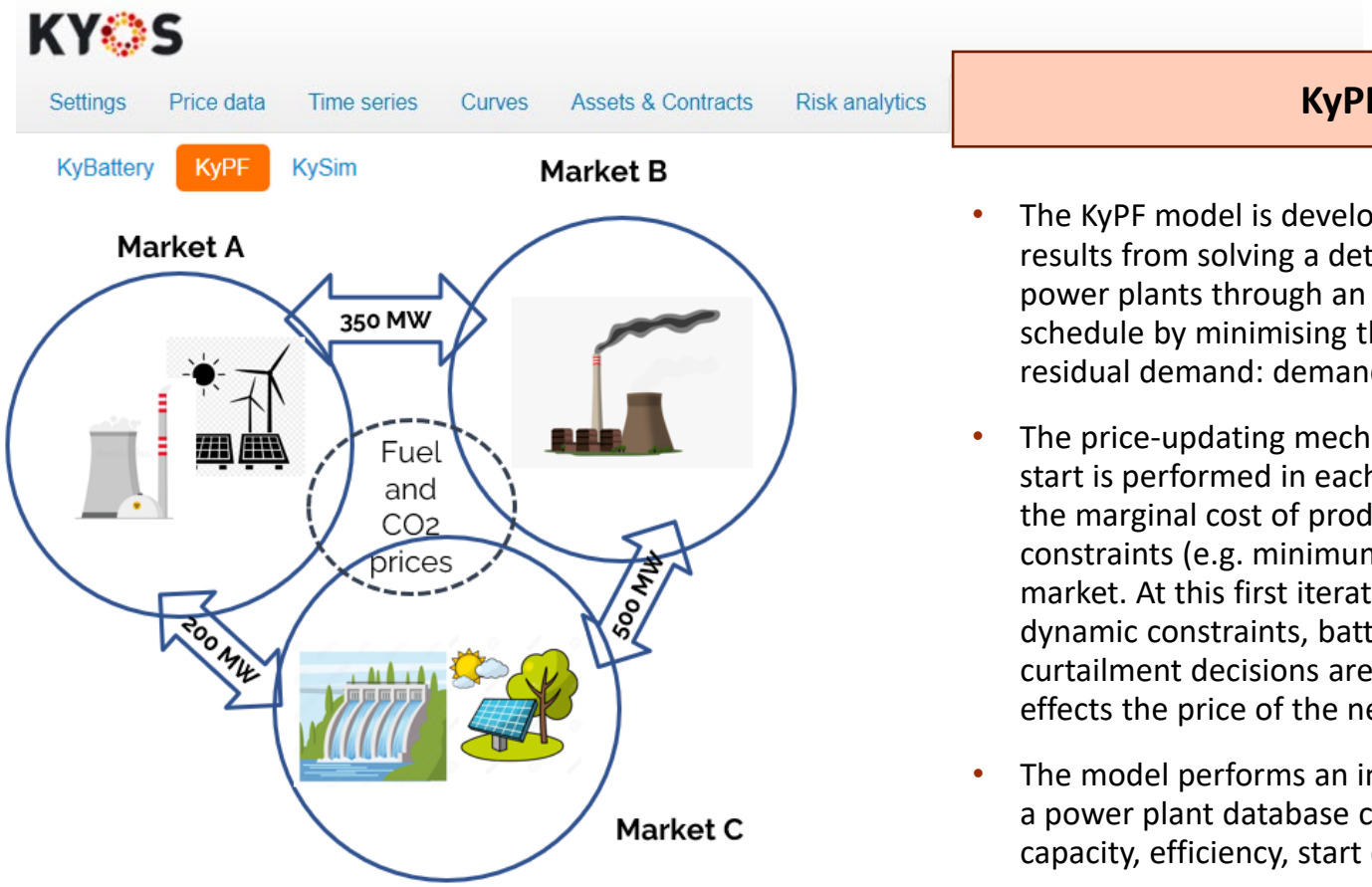
Annex 4 – KYOS fundamental power market model “KyPF”



KyPF: fundamental power market model

- The KyPF model is developed for the construction of hourly forward power price curves that results from solving a deterministic unit commitment problem for economic dispatch of power plants through an optimisation procedure. It determines the optimal power dispatch schedule by minimising the total cost to meet the power demand (subject in fact to the residual demand: demand minus solar+wind+hydro run-of-river).
- The price-updating mechanism behind KyPF is iterative. To initialize these iterations, a warm-start is performed in each market zone. A warm-start is a merit-order dispatch, in which only the marginal cost of production of each power plant plays a role, while their dynamic constraints (e.g. minimum on/off times, etc.) don’t. This forms a first power price vector per market. At this first iteration (and the next ones), power plants then do include their dynamic constraints, batteries & electrolysers are also dispatched, then renewable curtailment decisions are taken and finally power flow between market zones. This in turn effects the price of the next iteration.
- The model performs an individual optimization for each power plant units. We use as inputs a power plant database covering more than 2000 plants in Europe and including info such as capacity, efficiency, start costs, etc.

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Annex 5 – Validation of synthetic intraday price series with statistical measures

We evaluate the reasonableness of the ID price series with statistical measures against historical prices

- We computed different statistical measures for comparing the synthesized intraday time series (and its relation with day-ahead) with historical prices and to validate the quality of the ID price series.
- Figure 1: It shows the distribution of 2024 intraday prices (grey) against a synthetic ID price path, where DA-ID spread are randomly sampled in the 01-01-2024 to 31-12-2024 time window and pasted to 2024 DA prices. By definition, both distributions should fairly well overlap.
- Figure 2: It shows the distribution of 2024 intraday prices (grey) against a synthetic ID price path, where DA-ID spread are randomly sampled in the 01-01-2024 to 31-12-2024 time window and pasted to a single 2030 DA price simulation.
- Figure 3: The standard deviation of ID prices in 2024 is 107 EUR/MWh (red vertical line). The standard deviation of ID prices in 2030 for each ID price series is calculated and plotted in purple bars. The standard deviation of the median ID price serie is 108 EUR/MWh (grey vertical line), very much aligned with 2024 std.

Figure 1

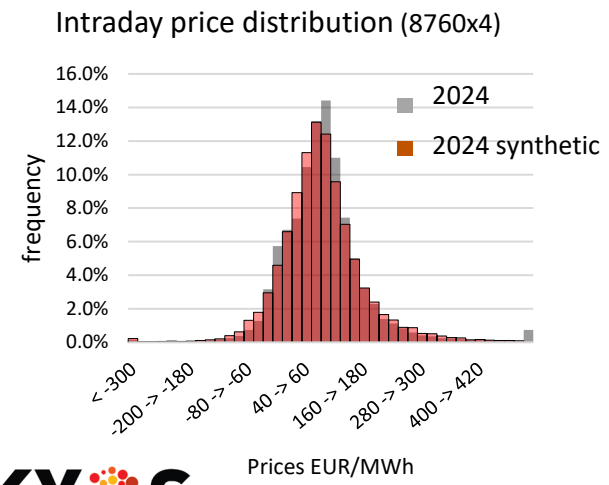


Figure 2

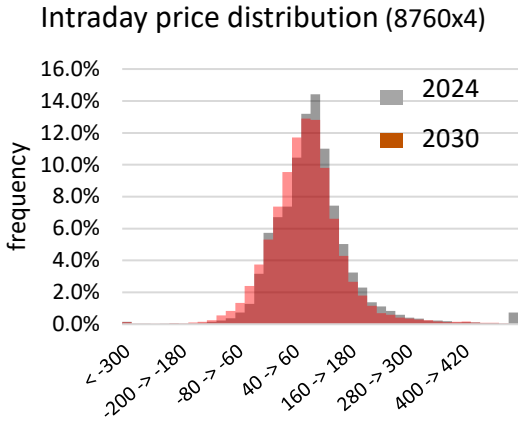
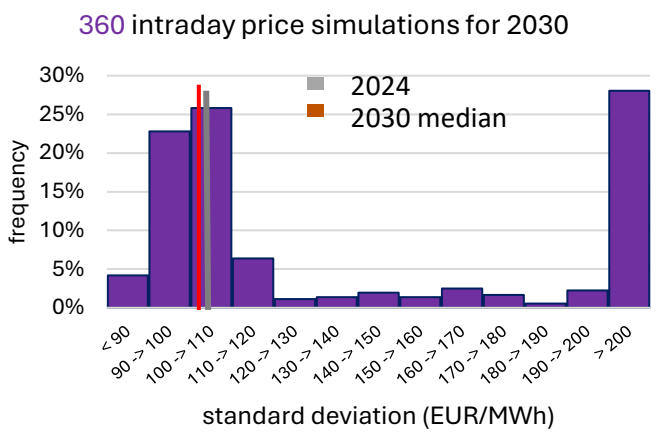


Figure 3



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Annex 6 - Perfect vs. Imperfect Foresight

Modelling approaches can explore how imperfect insight affects the value of BESS

- BESS optimizers do not have perfect foresight of future prices when making trading decisions. Real market revenues from participants are therefore likely to be lower than optimal perfect-foresight modeled revenues. To assess the importance of this in our modelling of BESS assets, we have used different modelling techniques to compare value under both perfect and foresight.
- The Least Squares Monte Carlo (LSMC) method uses Monte Carlo simulation to price American-style options (financial derivatives that can be exercised before expiration) by estimating continuation values through regression techniques. The method addresses the computational challenge of optimal exercise decisions without requiring perfect knowledge of future optimal strategies, but it does not specifically model imperfect foresight about future price forecasts.
- BESS operations involve continuous charge/discharge decisions that exhibit option-like characteristics. Every time period, operators face decisions about whether to charge, discharge, or remain idle based on current market conditions and future expectations. Such decisions are similar to American style options as bidding decisions in the intraday market are made prior to delivery time. LSMC methods can provide some insight into valuations under imperfect insight.
- KYOS modelled intraday trading revenues for a BESS to show only a slight drop of **4%** when optimized with the Least-Square Monte Carlo methods (**without perfect foresight** of prices), compared to the Dynamic programming optimization method (**with perfect foresight** of the prices).
- This relatively small difference in value reflects the computational approximation inherent in LSMC methods which might be thought as approximating the impact of estimation without perfect foresight, though this analogy has limitations since LSMC approximation errors can differ from real-world forecasting uncertainties.
- **We do not recommend making an adjustment for perfect foresight at this point as (1) the analysis is insufficiently robust and (2) it is small enough to be covered by overweighting the day-ahead market in revenue shares.**

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